```
OpenROAD Static Timing Analysis Report
Design: vsdbabysoc | Library: sky130_fd_sc_hd
SS corner: 18 ns clock, post-synthesis netlist, ideal clock, no wire parasitics (OpenSTA)

OpenROAD unknown
Features included (+) or not (-): -GPU +GUI -Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
[INFO ODB-0227] LEF file: /OpenROAD-flow-scripts/flow/platforms/sky130hd/lef/sky130_fd_sc_hd.tlef, created 13 layers, 25 vias
[INFO ODB-0227] LEF file: /OpenROAD-flow-scripts/flow/platforms/sky130hd/lef/sky130_fd_sc_hd_merged.lef, created 441 library cells
[WARNING ODB-0220] WARNING (LEFPARS-2008): NOWIREEXTENSIONATPIN statement is obsolete in version 5.6 or later.
The NOWIREEXTENSIONATPIN statement will be ignored. See file /OpenROAD-flow-scripts/flow/designs/sky130hd/vsdbabysoc/lef/avsdpll.lef at line 2.

[WARNING ODB-0186] macro avsdpll references unknown site unithddb1
[INFO ODB-0227] LEF file: /OpenROAD-flow-scripts/flow/designs/sky130hd/vsdbabysoc/lef/avsdpll.lef, created 1 library cells
[WARNING ODB-0220] WARNING (LEFPARS-2008): NOWIREEXTENSIONATPIN statement is obsolete in version 5.6 or later.
The NOWIREEXTENSIONATPIN statement will be ignored. See file /OpenROAD-flow-scripts/flow/designs/sky130hd/vsdbabysoc/lef/avsddac.lef at line 2.

[INFO ODB-0227] LEF file: /OpenROAD-flow-scripts/flow/designs/sky130hd/vsdbabysoc/lef/avsddac.lef, created 1 library cells
========== SS SETUP ==========
Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][30]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   2.60    2.60 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.18    3.78 ^ _11021_/SUM (sky130_fd_sc_hd__ha_1)
   0.75    4.53 ^ _05700_/X (sky130_fd_sc_hd__clkbuf_1)
   0.49    5.02 v _05701_/Y (sky130_fd_sc_hd__nand2_1)
   2.72    7.74 v _05703_/X (sky130_fd_sc_hd__or4_1)
   2.46   10.20 v _05704_/X (sky130_fd_sc_hd__or4_1)
   0.80   10.99 ^ _05705_/Y (sky130_fd_sc_hd__nor3_1)
   0.47   11.46 v _05716_/Y (sky130_fd_sc_hd__o31ai_1)
   0.56   12.02 v _05718_/X (sky130_fd_sc_hd__and2_1)
   0.65   12.67 v _09638_/X (sky130_fd_sc_hd__a21o_1)
   0.81   13.48 ^ _09640_/Y (sky130_fd_sc_hd__o221ai_1)
   1.00   14.48 ^ _09641_/X (sky130_fd_sc_hd__clkbuf_1)
   0.44   14.92 v _09828_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00   14.92 v core.CPU_Xreg_value_a4[2][30]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.92   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[2][30]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.85   17.15   library setup time
          17.15   data required time
---------------------------------------------------------
          17.15   data required time
         -14.92   data arrival time
---------------------------------------------------------
           2.22   slack (MET)


Startpoint: core.CPU_imm_a3[30]$_DFF_P_
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           (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[8][30]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   2.60    2.60 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.18    3.78 ^ _11021_/SUM (sky130_fd_sc_hd__ha_1)
   0.75    4.53 ^ _05700_/X (sky130_fd_sc_hd__clkbuf_1)
   0.49    5.02 v _05701_/Y (sky130_fd_sc_hd__nand2_1)
   2.72    7.74 v _05703_/X (sky130_fd_sc_hd__or4_1)
   2.46   10.20 v _05704_/X (sky130_fd_sc_hd__or4_1)
   0.80   10.99 ^ _05705_/Y (sky130_fd_sc_hd__nor3_1)
   0.47   11.46 v _05716_/Y (sky130_fd_sc_hd__o31ai_1)
   0.56   12.02 v _05718_/X (sky130_fd_sc_hd__and2_1)
   0.65   12.67 v _09638_/X (sky130_fd_sc_hd__a21o_1)
   0.81   13.48 ^ _09640_/Y (sky130_fd_sc_hd__o221ai_1)
   1.00   14.48 ^ _09641_/X (sky130_fd_sc_hd__clkbuf_1)
   0.44   14.92 v _10042_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00   14.92 v core.CPU_Xreg_value_a4[8][30]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.92   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[8][30]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.85   17.15   library setup time
          17.15   data required time
---------------------------------------------------------
          17.15   data required time
         -14.92   data arrival time
---------------------------------------------------------
           2.22   slack (MET)


Startpoint: core.CPU_imm_a3[30]$_DFF_P_
           (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[3][30]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   2.60    2.60 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.18    3.78 ^ _11021_/SUM (sky130_fd_sc_hd__ha_1)
   0.75    4.53 ^ _05700_/X (sky130_fd_sc_hd__clkbuf_1)
   0.49    5.02 v _05701_/Y (sky130_fd_sc_hd__nand2_1)
   2.72    7.74 v _05703_/X (sky130_fd_sc_hd__or4_1)
   2.46   10.20 v _05704_/X (sky130_fd_sc_hd__or4_1)
   0.80   10.99 ^ _05705_/Y (sky130_fd_sc_hd__nor3_1)
   0.47   11.46 v _05716_/Y (sky130_fd_sc_hd__o31ai_1)
   0.56   12.02 v _05718_/X (sky130_fd_sc_hd__and2_1)
   0.65   12.67 v _09638_/X (sky130_fd_sc_hd__a21o_1)
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   0.81   13.48 ^ _09640_/Y (sky130_fd_sc_hd__o221ai_1)
   1.00   14.48 ^ _09641_/X (sky130_fd_sc_hd__clkbuf_1)
   0.44   14.92 v _09942_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00   14.92 v core.CPU_Xreg_value_a4[3][30]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.92   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[3][30]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.85   17.15   library setup time
          17.15   data required time
-----------------------------------------------------------
          17.15   data required time
         -14.92   data arrival time
-----------------------------------------------------------
           2.22   slack (MET)


Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[11][30]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   2.60    2.60 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.18    3.78 ^ _11021_/SUM (sky130_fd_sc_hd__ha_1)
   0.75    4.53 ^ _05700_/X (sky130_fd_sc_hd__clkbuf_1)
   0.49    5.02 v _05701_/Y (sky130_fd_sc_hd__nand2_1)
   2.72    7.74 v _05703_/X (sky130_fd_sc_hd__or4_1)
   2.46   10.20 v _05704_/X (sky130_fd_sc_hd__or4_1)
   0.80   10.99 ^ _05705_/Y (sky130_fd_sc_hd__nor3_1)
   0.47   11.46 v _05716_/Y (sky130_fd_sc_hd__o31ai_1)
   0.56   12.02 v _05718_/X (sky130_fd_sc_hd__and2_1)
   0.65   12.67 v _09638_/X (sky130_fd_sc_hd__a21o_1)
   0.81   13.48 ^ _09640_/Y (sky130_fd_sc_hd__o221ai_1)
   1.00   14.48 ^ _09641_/X (sky130_fd_sc_hd__clkbuf_1)
   0.44   14.92 v _10316_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00   14.92 v core.CPU_Xreg_value_a4[11][30]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.92   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[11][30]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.85   17.15   library setup time
          17.15   data required time
---------------------------------------------------------
          17.15   data required time
         -14.92   data arrival time
---------------------------------------------------------
           2.22   slack (MET)


Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
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Endpoint: core.CPU_Xreg_value_a4[9][30]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   2.60    2.60 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.18    3.78 ^ _11021_/SUM (sky130_fd_sc_hd__ha_1)
   0.75    4.53 ^ _05700_/X (sky130_fd_sc_hd__clkbuf_1)
   0.49    5.02 v _05701_/Y (sky130_fd_sc_hd__nand2_1)
   2.72    7.74 v _05703_/X (sky130_fd_sc_hd__or4_1)
   2.46   10.20 v _05704_/X (sky130_fd_sc_hd__or4_1)
   0.80   10.99 ^ _05705_/Y (sky130_fd_sc_hd__nor3_1)
   0.47   11.46 v _05716_/Y (sky130_fd_sc_hd__o31ai_1)
   0.56   12.02 v _05718_/X (sky130_fd_sc_hd__and2_1)
   0.65   12.67 v _09638_/X (sky130_fd_sc_hd__a21o_1)
   0.81   13.48 ^ _09640_/Y (sky130_fd_sc_hd__o221ai_1)
   1.00   14.48 ^ _09641_/X (sky130_fd_sc_hd__clkbuf_1)
   0.44   14.92 v _10135_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00   14.92 v core.CPU_Xreg_value_a4[9][30]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.92   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[9][30]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.85   17.15   library setup time
          17.15   data required time
---------------------------------------------------------
          17.15   data required time
         -14.92   data arrival time
---------------------------------------------------------
           2.22   slack (MET)


Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[10][30]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   2.60    2.60 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.18    3.78 ^ _11021_/SUM (sky130_fd_sc_hd__ha_1)
   0.75    4.53 ^ _05700_/X (sky130_fd_sc_hd__clkbuf_1)
   0.49    5.02 v _05701_/Y (sky130_fd_sc_hd__nand2_1)
   2.72    7.74 v _05703_/X (sky130_fd_sc_hd__or4_1)
   2.46   10.20 v _05704_/X (sky130_fd_sc_hd__or4_1)
   0.80   10.99 ^ _05705_/Y (sky130_fd_sc_hd__nor3_1)
   0.47   11.46 v _05716_/Y (sky130_fd_sc_hd__o31ai_1)
   0.56   12.02 v _05718_/X (sky130_fd_sc_hd__and2_1)
   0.65   12.67 v _09638_/X (sky130_fd_sc_hd__a21o_1)
   0.81   13.48 ^ _09640_/Y (sky130_fd_sc_hd__o221ai_1)
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   1.00   14.48 ^ _09641_/X (sky130_fd_sc_hd__clkbuf_1)
   0.44   14.92 v _10226_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00   14.92 v core.CPU_Xreg_value_a4[10][30]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.92   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[10][30]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.85   17.15   library setup time
          17.15   data required time
-----------------------------------------------------------
          17.15   data required time
         -14.92   data arrival time
-----------------------------------------------------------
           2.22   slack (MET)


Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][30]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   2.60    2.60 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.18    3.78 ^ _11021_/SUM (sky130_fd_sc_hd__ha_1)
   0.75    4.53 ^ _05700_/X (sky130_fd_sc_hd__clkbuf_1)
   0.49    5.02 v _05701_/Y (sky130_fd_sc_hd__nand2_1)
   2.72    7.74 v _05703_/X (sky130_fd_sc_hd__or4_1)
   2.46   10.20 v _05704_/X (sky130_fd_sc_hd__or4_1)
   0.80   10.99 ^ _05705_/Y (sky130_fd_sc_hd__nor3_1)
   0.47   11.46 v _05716_/Y (sky130_fd_sc_hd__o31ai_1)
   0.56   12.02 v _05718_/X (sky130_fd_sc_hd__and2_1)
   0.65   12.67 v _09638_/X (sky130_fd_sc_hd__a21o_1)
   0.81   13.48 ^ _09640_/Y (sky130_fd_sc_hd__o221ai_1)
   1.00   14.48 ^ _09641_/X (sky130_fd_sc_hd__clkbuf_1)
   0.44   14.92 v _10503_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00   14.92 v core.CPU_Xreg_value_a4[17][30]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.92   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[17][30]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.85   17.15   library setup time
          17.15   data required time
-----------------------------------------------------------
          17.15   data required time
         -14.92   data arrival time
-----------------------------------------------------------
           2.23   slack (MET)


Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[16][30]$_SDFFE_PP0P_
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```
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   2.60    2.60 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.18    3.78 ^ _11021_/SUM (sky130_fd_sc_hd__ha_1)
   0.75    4.53 ^ _05700_/X (sky130_fd_sc_hd__clkbuf_1)
   0.49    5.02 v _05701_/Y (sky130_fd_sc_hd__nand2_1)
   2.72    7.74 v _05703_/X (sky130_fd_sc_hd__or4_1)
   2.46   10.20 v _05704_/X (sky130_fd_sc_hd__or4_1)
   0.80   10.99 ^ _05705_/Y (sky130_fd_sc_hd__nor3_1)
   0.47   11.46 v _05716_/Y (sky130_fd_sc_hd__o31ai_1)
   0.56   12.02 v _05718_/X (sky130_fd_sc_hd__and2_1)
   0.65   12.67 v _09638_/X (sky130_fd_sc_hd__a21o_1)
   0.81   13.48 ^ _09640_/Y (sky130_fd_sc_hd__o221ai_1)
   1.00   14.48 ^ _09641_/X (sky130_fd_sc_hd__clkbuf_1)
   0.44   14.92 v _10412_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00   14.92 v core.CPU_Xreg_value_a4[16][30]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.92   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[16][30]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.85   17.15   library setup time
          17.15   data required time
---------------------------------------------------------
          17.15   data required time
         -14.92   data arrival time
---------------------------------------------------------
           2.23   slack (MET)


Startpoint: core.CPU_imm_a3[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[1][30]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   2.60    2.60 ^ core.CPU_imm_a3[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.18    3.78 ^ _11021_/SUM (sky130_fd_sc_hd__ha_1)
   0.75    4.53 ^ _05700_/X (sky130_fd_sc_hd__clkbuf_1)
   0.49    5.02 v _05701_/Y (sky130_fd_sc_hd__nand2_1)
   2.72    7.74 v _05703_/X (sky130_fd_sc_hd__or4_1)
   2.46   10.20 v _05704_/X (sky130_fd_sc_hd__or4_1)
   0.80   10.99 ^ _05705_/Y (sky130_fd_sc_hd__nor3_1)
   0.47   11.46 v _05716_/Y (sky130_fd_sc_hd__o31ai_1)
   0.56   12.02 v _05718_/X (sky130_fd_sc_hd__and2_1)
   0.65   12.67 v _09638_/X (sky130_fd_sc_hd__a21o_1)
   0.81   13.48 ^ _09640_/Y (sky130_fd_sc_hd__o221ai_1)
   1.00   14.48 ^ _09641_/X (sky130_fd_sc_hd__clkbuf_1)
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   0.41   14.89 v _09648_/Y (sky130_fd_sc_hd__o21ai_0)
   0.00   14.89 v core.CPU_Xreg_value_a4[1][30]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.89   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[1][30]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.86   17.14   library setup time
          17.14   data required time
-----------------------------------------------------------
          17.14   data required time
         -14.89   data arrival time
-----------------------------------------------------------
           2.25   slack (MET)


Startpoint: core.CPU_src2_value_a3[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[10][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.53    1.53 ^ core.CPU_src2_value_a3[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.23    2.76 ^ _05848_/X (sky130_fd_sc_hd__clkbuf_1)
   0.57    3.33 v _07825_/Y (sky130_fd_sc_hd__inv_1)
   1.54    4.87 v _11038_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    5.56 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.48    6.03 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   2.56    8.60 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   1.25    9.84 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.70   10.54 v _06201_/X (sky130_fd_sc_hd__and2_1)
   1.19   11.73 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.58   12.31 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   1.19   13.50 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.94   14.45 ^ _06374_/X (sky130_fd_sc_hd__buf_2)
   0.33   14.78 v _10241_/Y (sky130_fd_sc_hd__o221ai_1)
   0.00   14.78 v core.CPU_Xreg_value_a4[10][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.78   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[10][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.87   17.13   library setup time
          17.13   data required time
-----------------------------------------------------------
          17.13   data required time
         -14.78   data arrival time
-----------------------------------------------------------
           2.35   slack (MET)


Startpoint: core.CPU_src2_value_a3[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[11][26]$_SDFFE_PP0P_
```

```
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.53    1.53 ^ core.CPU_src2_value_a3[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.23    2.76 ^ _05848_/X (sky130_fd_sc_hd__clkbuf_1)
   0.57    3.33 v _07825_/Y (sky130_fd_sc_hd__inv_1)
   1.54    4.87 v _11038_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    5.56 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.48    6.03 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   2.56    8.60 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   1.25    9.84 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.70   10.54 v _06201_/X (sky130_fd_sc_hd__and2_1)
   1.19   11.73 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.58   12.31 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   1.19   13.50 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.94   14.45 ^ _06374_/X (sky130_fd_sc_hd__buf_2)
   0.33   14.78 v _10333_/Y (sky130_fd_sc_hd__o221ai_1)
   0.00   14.78 v core.CPU_Xreg_value_a4[11][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.78   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[11][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.87   17.13   library setup time
          17.13   data required time
---------------------------------------------------------
          17.13   data required time
         -14.78   data arrival time
---------------------------------------------------------
           2.35   slack (MET)


Startpoint: core.CPU_src2_value_a3[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[16][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.53    1.53 ^ core.CPU_src2_value_a3[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.23    2.76 ^ _05848_/X (sky130_fd_sc_hd__clkbuf_1)
   0.57    3.33 v _07825_/Y (sky130_fd_sc_hd__inv_1)
   1.54    4.87 v _11038_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    5.56 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.48    6.03 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   2.56    8.60 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   1.25    9.84 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.70   10.54 v _06201_/X (sky130_fd_sc_hd__and2_1)
   1.19   11.73 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.58   12.31 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
```

```
   1.19   13.50 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.94   14.45 ^ _06374_/X (sky130_fd_sc_hd__buf_2)
   0.33   14.78 v _10428_/Y (sky130_fd_sc_hd__o221ai_1)
   0.00   14.78 v core.CPU_Xreg_value_a4[16][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.78   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[16][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.87   17.13   library setup time
          17.13   data required time
-----------------------------------------------------------
          17.13   data required time
         -14.78   data arrival time
-----------------------------------------------------------
           2.35   slack (MET)


Startpoint: core.CPU_src2_value_a3[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.53    1.53 ^ core.CPU_src2_value_a3[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.23    2.76 ^ _05848_/X (sky130_fd_sc_hd__clkbuf_1)
   0.57    3.33 v _07825_/Y (sky130_fd_sc_hd__inv_1)
   1.54    4.87 v _11038_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    5.56 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.48    6.03 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   2.56    8.60 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   1.25    9.84 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.70   10.54 v _06201_/X (sky130_fd_sc_hd__and2_1)
   1.19   11.73 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.58   12.31 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   1.19   13.50 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.94   14.45 ^ _06374_/X (sky130_fd_sc_hd__buf_2)
   0.33   14.78 v _10518_/Y (sky130_fd_sc_hd__o221ai_1)
   0.00   14.78 v core.CPU_Xreg_value_a4[17][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.78   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[17][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.87   17.13   library setup time
          17.13   data required time
-----------------------------------------------------------
          17.13   data required time
         -14.78   data arrival time
-----------------------------------------------------------
           2.35   slack (MET)


Startpoint: core.CPU_src2_value_a3[2]$_DFF_P_
```

```
           (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.53    1.53 ^ core.CPU_src2_value_a3[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.23    2.76 ^ _05848_/X (sky130_fd_sc_hd__clkbuf_1)
   0.57    3.33 v _07825_/Y (sky130_fd_sc_hd__inv_1)
   1.54    4.87 v _11038_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    5.56 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.48    6.03 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   2.56    8.60 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   1.25    9.84 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.70   10.54 v _06201_/X (sky130_fd_sc_hd__and2_1)
   1.19   11.73 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.58   12.31 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   1.19   13.50 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.94   14.45 ^ _06374_/X (sky130_fd_sc_hd__buf_2)
   0.33   14.78 v _09847_/Y (sky130_fd_sc_hd__o221ai_1)
   0.00   14.78 v core.CPU_Xreg_value_a4[2][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.78   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[2][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.87   17.13   library setup time
          17.13   data required time
---------------------------------------------------------
          17.13   data required time
         -14.78   data arrival time
---------------------------------------------------------
           2.35   slack (MET)


Startpoint: core.CPU_src2_value_a3[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[3][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.53    1.53 ^ core.CPU_src2_value_a3[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.23    2.76 ^ _05848_/X (sky130_fd_sc_hd__clkbuf_1)
   0.57    3.33 v _07825_/Y (sky130_fd_sc_hd__inv_1)
   1.54    4.87 v _11038_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    5.56 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.48    6.03 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   2.56    8.60 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   1.25    9.84 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.70   10.54 v _06201_/X (sky130_fd_sc_hd__and2_1)
```

```
   1.19   11.73 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.58   12.31 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   1.19   13.50 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.94   14.45 ^ _06374_/X (sky130_fd_sc_hd__buf_2)
   0.33   14.78 v _09959_/Y (sky130_fd_sc_hd__o221ai_1)
   0.00   14.78 v core.CPU_Xreg_value_a4[3][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.78   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[3][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.87   17.13   library setup time
          17.13   data required time
-----------------------------------------------------------
          17.13   data required time
         -14.78   data arrival time
-----------------------------------------------------------
           2.35   slack (MET)


Startpoint: core.CPU_src2_value_a3[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[8][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.53    1.53 ^ core.CPU_src2_value_a3[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.23    2.76 ^ _05848_/X (sky130_fd_sc_hd__clkbuf_1)
   0.57    3.33 v _07825_/Y (sky130_fd_sc_hd__inv_1)
   1.54    4.87 v _11038_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    5.56 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.48    6.03 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   2.56    8.60 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   1.25    9.84 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.70   10.54 v _06201_/X (sky130_fd_sc_hd__and2_1)
   1.19   11.73 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.58   12.31 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   1.19   13.50 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.94   14.45 ^ _06374_/X (sky130_fd_sc_hd__buf_2)
   0.33   14.78 v _10060_/Y (sky130_fd_sc_hd__o221ai_1)
   0.00   14.78 v core.CPU_Xreg_value_a4[8][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.78   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[8][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.87   17.13   library setup time
          17.13   data required time
-----------------------------------------------------------
          17.13   data required time
         -14.78   data arrival time
-----------------------------------------------------------
           2.35   slack (MET)
```

```
Startpoint: core.CPU_src2_value_a3[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[9][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.53    1.53 ^ core.CPU_src2_value_a3[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.23    2.76 ^ _05848_/X (sky130_fd_sc_hd__clkbuf_1)
   0.57    3.33 v _07825_/Y (sky130_fd_sc_hd__inv_1)
   1.54    4.87 v _11038_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    5.56 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.48    6.03 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   2.56    8.60 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   1.25    9.84 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.70   10.54 v _06201_/X (sky130_fd_sc_hd__and2_1)
   1.19   11.73 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.58   12.31 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   1.19   13.50 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.94   14.45 ^ _06374_/X (sky130_fd_sc_hd__buf_2)
   0.33   14.78 v _10151_/Y (sky130_fd_sc_hd__o221ai_1)
   0.00   14.78 v core.CPU_Xreg_value_a4[9][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.78   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[9][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.87   17.13   library setup time
          17.13   data required time
---------------------------------------------------------
          17.13   data required time
         -14.78   data arrival time
---------------------------------------------------------
           2.35   slack (MET)


Startpoint: core.CPU_src2_value_a3[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_src1_value_a3[26]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.53    1.53 ^ core.CPU_src2_value_a3[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.23    2.76 ^ _05848_/X (sky130_fd_sc_hd__clkbuf_1)
   0.57    3.33 v _07825_/Y (sky130_fd_sc_hd__inv_1)
   1.54    4.87 v _11038_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    5.56 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.48    6.03 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   2.56    8.60 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
```

```
   1.25    9.84 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.70   10.54 v _06201_/X (sky130_fd_sc_hd__and2_1)
   1.19   11.73 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.58   12.31 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   1.19   13.50 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.94   14.45 ^ _06374_/X (sky130_fd_sc_hd__buf_2)
   0.26   14.71 v _06375_/Y (sky130_fd_sc_hd__mux2i_1)
   0.00   14.71 v core.CPU_src1_value_a3[26]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.71   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_src1_value_a3[26]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.88   17.12   library setup time
          17.12   data required time
-----------------------------------------------------------
          17.12   data required time
         -14.71   data arrival time
-----------------------------------------------------------
           2.41   slack (MET)


Startpoint: core.CPU_src2_value_a3[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[10][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.53    1.53 ^ core.CPU_src2_value_a3[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.23    2.76 ^ _05848_/X (sky130_fd_sc_hd__clkbuf_1)
   0.57    3.33 v _07825_/Y (sky130_fd_sc_hd__inv_1)
   1.54    4.87 v _11038_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    5.56 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.48    6.03 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   2.56    8.60 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   1.25    9.84 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.70   10.54 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.76   11.30 ^ _06206_/Y (sky130_fd_sc_hd__o21ai_0)
   0.40   11.70 v _06207_/Y (sky130_fd_sc_hd__xnor2_1)
   0.84   12.54 ^ _06208_/Y (sky130_fd_sc_hd__a22oi_1)
   1.28   13.82 ^ _06209_/X (sky130_fd_sc_hd__clkbuf_1)
   0.59   14.41 v _10236_/Y (sky130_fd_sc_hd__a311oi_1)
   0.00   14.41 v core.CPU_Xreg_value_a4[10][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.41   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[10][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.87   17.13   library setup time
          17.13   data required time
---------------------------------------------------------
          17.13   data required time
         -14.41   data arrival time
---------------------------------------------------------
```

```
           2.72   slack (MET)


Startpoint: core.CPU_src2_value_a3[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[11][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.53    1.53 ^ core.CPU_src2_value_a3[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.23    2.76 ^ _05848_/X (sky130_fd_sc_hd__clkbuf_1)
   0.57    3.33 v _07825_/Y (sky130_fd_sc_hd__inv_1)
   1.54    4.87 v _11038_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    5.56 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.48    6.03 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   2.56    8.60 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   1.25    9.84 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.70   10.54 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.76   11.30 ^ _06206_/Y (sky130_fd_sc_hd__o21ai_0)
   0.40   11.70 v _06207_/Y (sky130_fd_sc_hd__xnor2_1)
   0.84   12.54 ^ _06208_/Y (sky130_fd_sc_hd__a22oi_1)
   1.28   13.82 ^ _06209_/X (sky130_fd_sc_hd__clkbuf_1)
   0.59   14.41 v _10326_/Y (sky130_fd_sc_hd__a311oi_1)
   0.00   14.41 v core.CPU_Xreg_value_a4[11][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.41   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[11][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.87   17.13   library setup time
          17.13   data required time
---------------------------------------------------------
          17.13   data required time
         -14.41   data arrival time
---------------------------------------------------------
           2.72   slack (MET)


Startpoint: core.CPU_src2_value_a3[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[16][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.53    1.53 ^ core.CPU_src2_value_a3[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.23    2.76 ^ _05848_/X (sky130_fd_sc_hd__clkbuf_1)
   0.57    3.33 v _07825_/Y (sky130_fd_sc_hd__inv_1)
   1.54    4.87 v _11038_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    5.56 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
```

```
   0.48    6.03 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   2.56    8.60 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   1.25    9.84 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.70   10.54 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.76   11.30 ^ _06206_/Y (sky130_fd_sc_hd__o21ai_0)
   0.40   11.70 v _06207_/Y (sky130_fd_sc_hd__xnor2_1)
   0.84   12.54 ^ _06208_/Y (sky130_fd_sc_hd__a22oi_1)
   1.28   13.82 ^ _06209_/X (sky130_fd_sc_hd__clkbuf_1)
   0.59   14.41 v _10423_/Y (sky130_fd_sc_hd__a311oi_1)
   0.00   14.41 v core.CPU_Xreg_value_a4[16][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.41   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[16][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.87   17.13   library setup time
          17.13   data required time
-----------------------------------------------------------
          17.13   data required time
         -14.41   data arrival time
-----------------------------------------------------------
           2.72   slack (MET)


Startpoint: core.CPU_src2_value_a3[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[1][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.53    1.53 ^ core.CPU_src2_value_a3[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.23    2.76 ^ _05848_/X (sky130_fd_sc_hd__clkbuf_1)
   0.57    3.33 v _07825_/Y (sky130_fd_sc_hd__inv_1)
   1.54    4.87 v _11038_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    5.56 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.48    6.03 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   2.56    8.60 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   1.25    9.84 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.70   10.54 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.76   11.30 ^ _06206_/Y (sky130_fd_sc_hd__o21ai_0)
   0.40   11.70 v _06207_/Y (sky130_fd_sc_hd__xnor2_1)
   0.84   12.54 ^ _06208_/Y (sky130_fd_sc_hd__a22oi_1)
   1.28   13.82 ^ _06209_/X (sky130_fd_sc_hd__clkbuf_1)
   0.59   14.41 v _09668_/Y (sky130_fd_sc_hd__a311oi_1)
   0.00   14.41 v core.CPU_Xreg_value_a4[1][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.41   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[1][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.87   17.13   library setup time
          17.13   data required time
---------------------------------------------------------
          17.13   data required time
```

```
         -14.41   data arrival time
-----------------------------------------------------------
           2.72   slack (MET)


Startpoint: core.CPU_src2_value_a3[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.53    1.53 ^ core.CPU_src2_value_a3[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.23    2.76 ^ _05848_/X (sky130_fd_sc_hd__clkbuf_1)
   0.57    3.33 v _07825_/Y (sky130_fd_sc_hd__inv_1)
   1.54    4.87 v _11038_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    5.56 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.48    6.03 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   2.56    8.60 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   1.25    9.84 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.70   10.54 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.76   11.30 ^ _06206_/Y (sky130_fd_sc_hd__o21ai_0)
   0.40   11.70 v _06207_/Y (sky130_fd_sc_hd__xnor2_1)
   0.84   12.54 ^ _06208_/Y (sky130_fd_sc_hd__a22oi_1)
   1.28   13.82 ^ _06209_/X (sky130_fd_sc_hd__clkbuf_1)
   0.59   14.41 v _09841_/Y (sky130_fd_sc_hd__a311oi_1)
   0.00   14.41 v core.CPU_Xreg_value_a4[2][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.41   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[2][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.87   17.13   library setup time
          17.13   data required time
---------------------------------------------------------
          17.13   data required time
         -14.41   data arrival time
---------------------------------------------------------
           2.72   slack (MET)


Startpoint: core.CPU_src2_value_a3[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[3][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.53    1.53 ^ core.CPU_src2_value_a3[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.23    2.76 ^ _05848_/X (sky130_fd_sc_hd__clkbuf_1)
   0.57    3.33 v _07825_/Y (sky130_fd_sc_hd__inv_1)
```

```
   1.54    4.87 v _11038_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    5.56 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.48    6.03 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   2.56    8.60 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   1.25    9.84 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.70   10.54 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.76   11.30 ^ _06206_/Y (sky130_fd_sc_hd__o21ai_0)
   0.40   11.70 v _06207_/Y (sky130_fd_sc_hd__xnor2_1)
   0.84   12.54 ^ _06208_/Y (sky130_fd_sc_hd__a22oi_1)
   1.28   13.82 ^ _06209_/X (sky130_fd_sc_hd__clkbuf_1)
   0.59   14.41 v _09953_/Y (sky130_fd_sc_hd__a311oi_1)
   0.00   14.41 v core.CPU_Xreg_value_a4[3][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.41   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[3][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.87   17.13   library setup time
          17.13   data required time
-----------------------------------------------------------
          17.13   data required time
         -14.41   data arrival time
-----------------------------------------------------------
           2.72   slack (MET)


Startpoint: core.CPU_src2_value_a3[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[8][28]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.53    1.53 ^ core.CPU_src2_value_a3[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.23    2.76 ^ _05848_/X (sky130_fd_sc_hd__clkbuf_1)
   0.57    3.33 v _07825_/Y (sky130_fd_sc_hd__inv_1)
   1.54    4.87 v _11038_/SUM (sky130_fd_sc_hd__ha_1)
   0.68    5.56 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.48    6.03 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   2.56    8.60 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   1.25    9.84 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.70   10.54 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.76   11.30 ^ _06206_/Y (sky130_fd_sc_hd__o21ai_0)
   0.40   11.70 v _06207_/Y (sky130_fd_sc_hd__xnor2_1)
   0.84   12.54 ^ _06208_/Y (sky130_fd_sc_hd__a22oi_1)
   1.28   13.82 ^ _06209_/X (sky130_fd_sc_hd__clkbuf_1)
   0.59   14.41 v _10055_/Y (sky130_fd_sc_hd__a311oi_1)
   0.00   14.41 v core.CPU_Xreg_value_a4[8][28]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          14.41   data arrival time

  18.00   18.00   clock clk (rise edge)
   0.00   18.00   clock network delay (ideal)
   0.00   18.00   clock reconvergence pessimism
          18.00 ^ core.CPU_Xreg_value_a4[8][28]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.87   17.13   library setup time
          17.13   data required time
```

```
-------------------------------------------------------------
         17.13   data required time
        -14.41   data arrival time
-------------------------------------------------------------
          2.72   slack (MET)


========== SS HOLD ==========
Startpoint: core.CPU_imm_a2[10]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_imm_a3[10]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_imm_a2[10]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_imm_a2[10]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_imm_a3[10]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_imm_a3[10]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_imm_a2[11]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_imm_a3[11]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_imm_a2[11]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_imm_a2[11]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_imm_a3[11]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_imm_a3[11]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
```

```
-------------------------------------------------------------
          1.12   slack (MET)


Startpoint: core.CPU_imm_a2[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_imm_a3[30]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_imm_a2[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_imm_a2[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_imm_a3[30]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_inc_pc_a2[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[2]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_inc_pc_a2[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_inc_pc_a2[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_inc_pc_a3[2]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_inc_pc_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)
```

```
Startpoint: core.CPU_inc_pc_a2[3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[3]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_inc_pc_a2[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_inc_pc_a2[3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_inc_pc_a3[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_inc_pc_a3[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_inc_pc_a2[4]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[4]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_inc_pc_a2[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_inc_pc_a2[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_inc_pc_a3[4]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_inc_pc_a3[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_inc_pc_a2[5]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[5]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
```

```
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_inc_pc_a2[5]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_inc_pc_a2[5]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_inc_pc_a3[5]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_inc_pc_a3[5]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_is_add_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_add_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_add_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_is_add_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_is_add_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_is_add_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_is_addi_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_addi_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
```

```
-------------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_addi_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_is_addi_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_is_addi_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_is_addi_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
-------------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
-------------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_is_beq_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_beq_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
-------------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_beq_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_is_beq_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_is_beq_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_is_beq_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
-------------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
-------------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_is_bne_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_bne_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
-------------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_bne_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
```

```
   0.91    0.91 v core.CPU_is_bne_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_is_bne_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_is_bne_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
-----------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
-----------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_is_load_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_load_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_load_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_is_load_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_is_load_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_is_load_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
-----------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
-----------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_is_s_instr_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_s_instr_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_s_instr_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_is_s_instr_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_is_s_instr_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time
```

```
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_is_s_instr_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
-----------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
-----------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_is_s_instr_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_s_instr_a4$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_s_instr_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_is_s_instr_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_is_s_instr_a4$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_is_s_instr_a4$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
-----------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
-----------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_is_sll_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_sll_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_sll_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_is_sll_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_is_sll_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_is_sll_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
```

```
  -0.22   -0.22   library hold time
          -0.22   data required time
-----------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
-----------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_is_slli_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_slli_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_slli_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_is_slli_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_is_slli_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_is_sub_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_sub_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_sub_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_is_sub_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_is_sub_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_is_sub_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
```

```
          -0.91   data arrival time
-------------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_rd_a2[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[0]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_rd_a2[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_rd_a2[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_rd_a3[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_rd_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_rd_a2[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[1]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_rd_a2[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_rd_a2[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_rd_a3[1]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_rd_a3[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)
```

```
Startpoint: core.CPU_rd_a2[3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[3]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_rd_a2[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_rd_a2[3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_rd_a3[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_rd_a3[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_rd_a2[4]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[4]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_rd_a2[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_rd_a2[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_rd_a3[4]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_rd_a3[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_rd_a4[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a5[0]$_DFF_P_
```

```
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_rd_a4[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_rd_a4[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_rd_a5[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_rd_a5[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_rd_a4[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a5[1]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_rd_a4[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_rd_a4[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_rd_a5[1]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_rd_a5[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_rd_a4[3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
```

```
  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_rd_a4[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_rd_a4[3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_rd_a5[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_rd_a5[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)


Startpoint: core.CPU_rd_a4[4]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a5[4]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_rd_a4[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.91    0.91 v core.CPU_rd_a4[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.91 v core.CPU_rd_a5[4]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
           0.91   data arrival time

   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00   clock reconvergence pessimism
           0.00 ^ core.CPU_rd_a5[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.22   -0.22   library hold time
          -0.22   data required time
---------------------------------------------------------
          -0.22   data required time
          -0.91   data arrival time
---------------------------------------------------------
           1.12   slack (MET)


========== SS WNS/TNS ==========
worst slack max 2.22
worst slack min 1.12
tns max 0.00
```

```
OpenROAD Static Timing Analysis Report
Design: vsdbabysoc | Library: sky130_fd_sc_hd
TT corner: design SDC clock, post-global-route (5_1_grt.odb), propagated clock

OpenROAD unknown
Features included (+) or not (-): -GPU +GUI -Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
========== TT SETUP ==========
Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[8][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
           0.07     0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13     0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36     0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16     0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16     0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36     0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15     0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97     0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25     0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14     0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.44     0.45    4.86 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.13     0.18    5.04 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.07    1.25     1.05    6.09 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.01    0.23     0.28    6.37 v _10060_/Y (sky130_fd_sc_hd__o221ai_1)
           0.23     0.00    6.37 v core.CPU_Xreg_value_a4[8][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                            6.37   data arrival time

                   11.00   11.00   clock clk (rise edge)
                    0.85   11.85   clock network delay (propagated)
                    0.00   11.85   clock reconvergence pessimism
                           11.85 ^ core.CPU_Xreg_value_a4[8][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.19   11.67   library setup time
                           11.67   data required time
---------------------------------------------------------------------
                           11.67   data required time
                           -6.37   data arrival time
---------------------------------------------------------------------
                            5.30   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
           0.07     0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
```

```
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.44    0.45    4.86 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.13    0.18    5.04 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.07    1.25    1.05    6.09 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.22    0.27    6.36 v _10706_/Y (sky130_fd_sc_hd__o221ai_1)
           0.22    0.00    6.36 v core.CPU_Xreg_value_a4[27][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.36   data arrival time

                   11.00   11.00   clock clk (rise edge)
                    0.85   11.85   clock network delay (propagated)
                    0.00   11.85   clock reconvergence pessimism
                           11.85 ^ core.CPU_Xreg_value_a4[27][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.18   11.67   library setup time
                           11.67   data required time
---------------------------------------------------------------------------
                           11.67   data required time
                           -6.36   data arrival time
---------------------------------------------------------------------------
                            5.31   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[1][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
           0.07     0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.44    0.45    4.86 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.13    0.18    5.04 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.07    1.25    1.05    6.09 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.19    0.29    6.38 v _09682_/Y (sky130_fd_sc_hd__o221ai_1)
           0.19    0.00    6.38 v core.CPU_Xreg_value_a4[1][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.38   data arrival time

                   11.00   11.00   clock clk (rise edge)
                    0.85   11.85   clock network delay (propagated)
                    0.00   11.85   clock reconvergence pessimism
                           11.85 ^ core.CPU_Xreg_value_a4[1][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.17   11.69   library setup time
```

```
                          11.69   data required time
-----------------------------------------------------------------------------
                          11.69   data required time
                          -6.38   data arrival time
-----------------------------------------------------------------------------
                           5.31   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.44    0.45    4.86 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.13    0.18    5.04 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.07    1.25    1.05    6.09 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.22    0.26    6.35 v _10518_/Y (sky130_fd_sc_hd__o221ai_1)
           0.22    0.00    6.35 v core.CPU_Xreg_value_a4[17][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.35   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.85   11.85   clock network delay (propagated)
                   0.00   11.85   clock reconvergence pessimism
                          11.85 ^ core.CPU_Xreg_value_a4[17][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.18   11.67   library setup time
                          11.67   data required time
-----------------------------------------------------------------------------
                          11.67   data required time
                          -6.35   data arrival time
-----------------------------------------------------------------------------
                           5.32   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[26][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
```

```
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.44    0.45    4.86 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.13    0.18    5.04 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.07    1.25    1.05    6.09 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.21    0.25    6.34 v _10617_/Y (sky130_fd_sc_hd__o221ai_1)
           0.21    0.00    6.34 v core.CPU_Xreg_value_a4[26][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.34   data arrival time

                   11.00   11.00   clock clk (rise edge)
                    0.85   11.85   clock network delay (propagated)
                    0.00   11.85   clock reconvergence pessimism
                           11.85 ^ core.CPU_Xreg_value_a4[26][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.18   11.67   library setup time
                           11.67   data required time
-----------------------------------------------------------------------------
                           11.67   data required time
                           -6.34   data arrival time
-----------------------------------------------------------------------------
                            5.33   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[3][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
           0.07     0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13     0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36     0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16     0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16     0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36     0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15     0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97     0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25     0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14     0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.44     0.45    4.86 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.13     0.18    5.04 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.07    1.25     1.05    6.09 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.21     0.25    6.34 v _09959_/Y (sky130_fd_sc_hd__o221ai_1)
           0.21     0.00    6.34 v core.CPU_Xreg_value_a4[3][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                            6.34   data arrival time

                   11.00    11.00   clock clk (rise edge)
                    0.85    11.85   clock network delay (propagated)
                    0.00    11.85   clock reconvergence pessimism
                            11.85 ^ core.CPU_Xreg_value_a4[3][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.18    11.68   library setup time
                            11.68   data required time
```

```
------------------------------------------------------------------------
                         11.68   data required time
                         -6.34   data arrival time
------------------------------------------------------------------------
                          5.34   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.44    0.45    4.86 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.13    0.18    5.04 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.07    1.25    1.05    6.09 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.21    0.25    6.34 v _09847_/Y (sky130_fd_sc_hd__o221ai_1)
           0.21    0.00    6.34 v core.CPU_Xreg_value_a4[2][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.34   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.85   11.85   clock network delay (propagated)
                   0.00   11.85   clock reconvergence pessimism
                          11.85 ^ core.CPU_Xreg_value_a4[2][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.18   11.68   library setup time
                          11.68   data required time
------------------------------------------------------------------------
                         11.68   data required time
                         -6.34   data arrival time
------------------------------------------------------------------------
                          5.34   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[16][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
```

```
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.44    0.45    4.86 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.13    0.18    5.04 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.07    1.25    1.05    6.09 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.21    0.24    6.33 v _10428_/Y (sky130_fd_sc_hd__o221ai_1)
           0.21    0.00    6.33 v core.CPU_Xreg_value_a4[16][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.33   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.85   11.85   clock network delay (propagated)
                   0.00   11.85   clock reconvergence pessimism
                          11.85 ^ core.CPU_Xreg_value_a4[16][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.18   11.68   library setup time
                          11.68   data required time
-----------------------------------------------------------------------------
                          11.68   data required time
                          -6.33   data arrival time
-----------------------------------------------------------------------------
                           5.34   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[9][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.44    0.45    4.86 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.13    0.18    5.04 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.07    1.25    1.05    6.09 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.21    0.24    6.33 v _10151_/Y (sky130_fd_sc_hd__o221ai_1)
           0.21    0.00    6.33 v core.CPU_Xreg_value_a4[9][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.33   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.85   11.85   clock network delay (propagated)
                   0.00   11.85   clock reconvergence pessimism
                          11.85 ^ core.CPU_Xreg_value_a4[9][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.18   11.68   library setup time
                          11.68   data required time
-----------------------------------------------------------------------------
```

```
                          11.68   data required time
                          -6.33   data arrival time
-----------------------------------------------------------------------------
                           5.35   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[10][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
           0.07     0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13     0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36     0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16     0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16     0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36     0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15     0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97     0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25     0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14     0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.44     0.45    4.86 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.13     0.18    5.04 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.07    1.25     1.05    6.09 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.20     0.23    6.32 v _10241_/Y (sky130_fd_sc_hd__o221ai_1)
           0.20     0.00    6.32 v core.CPU_Xreg_value_a4[10][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                            6.32   data arrival time

                   11.00   11.00   clock clk (rise edge)
                    0.85   11.85   clock network delay (propagated)
                    0.00   11.85   clock reconvergence pessimism
                           11.85 ^ core.CPU_Xreg_value_a4[10][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.17   11.68   library setup time
                           11.68   data required time
---------------------------------------------------------------------------
                           11.68   data required time
                           -6.32   data arrival time
---------------------------------------------------------------------------
                            5.36   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_src2_value_a3[26]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
           0.07     0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13     0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36     0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16     0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
```

```
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.44    0.45    4.86 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.13    0.18    5.04 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.07    1.25    1.05    6.09 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.18    0.24    6.33 v _07135_/Y (sky130_fd_sc_hd__o21ai_0)
           0.18    0.00    6.33 v core.CPU_src2_value_a3[26]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.33   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.86   11.86   clock network delay (propagated)
                   0.00   11.86   clock reconvergence pessimism
                          11.86 ^ core.CPU_src2_value_a3[26]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.16   11.69   library setup time
                          11.69   data required time
-----------------------------------------------------------------------------
                          11.69   data required time
                          -6.33   data arrival time
-----------------------------------------------------------------------------
                           5.36   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[11][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.44    0.45    4.86 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.13    0.18    5.04 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.07    1.25    1.05    6.09 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.19    0.22    6.31 v _10333_/Y (sky130_fd_sc_hd__o221ai_1)
           0.19    0.00    6.31 v core.CPU_Xreg_value_a4[11][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.31   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.85   11.85   clock network delay (propagated)
                   0.00   11.85   clock reconvergence pessimism
                          11.85 ^ core.CPU_Xreg_value_a4[11][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.17   11.68   library setup time
                          11.68   data required time
-----------------------------------------------------------------------------
                          11.68   data required time
```

```
                          -6.31   data arrival time
-------------------------------------------------------------------------
                           5.37   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_src1_value_a3[26]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
           0.07     0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13     0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36     0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16     0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16     0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36     0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15     0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97     0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25     0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14     0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.44     0.45    4.86 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.13     0.18    5.04 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.07    1.25     1.05    6.09 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.21     0.20    6.29 v _06375_/Y (sky130_fd_sc_hd__mux2i_1)
           0.21     0.00    6.29 v core.CPU_src1_value_a3[26]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                            6.29   data arrival time

                   11.00   11.00   clock clk (rise edge)
                    0.97   11.97   clock network delay (propagated)
                    0.00   11.97   clock reconvergence pessimism
                           11.97 ^ core.CPU_src1_value_a3[26]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.18   11.79   library setup time
                           11.79   data required time
-------------------------------------------------------------------------
                           11.79   data required time
                           -6.29   data arrival time
-------------------------------------------------------------------------
                            5.50   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[26][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
           0.07     0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13     0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36     0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16     0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16     0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
```

```
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.29    0.30    4.71 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.12    0.13    4.85 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    1.16    0.98    5.82 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.01    0.22    0.30    6.13 v _10623_/Y (sky130_fd_sc_hd__o21ai_0)
           0.22    0.00    6.13 v core.CPU_Xreg_value_a4[26][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.13   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.85   11.85   clock network delay (propagated)
                   0.00   11.85   clock reconvergence pessimism
                          11.85 ^ core.CPU_Xreg_value_a4[26][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.18   11.67   library setup time
                          11.67   data required time
-----------------------------------------------------------------------------
                          11.67   data required time
                          -6.13   data arrival time
-----------------------------------------------------------------------------
                           5.54   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.29    0.30    4.71 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.12    0.13    4.85 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    1.16    0.98    5.82 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.20    0.28    6.10 v _09855_/Y (sky130_fd_sc_hd__o21ai_0)
           0.20    0.00    6.10 v core.CPU_Xreg_value_a4[2][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.10   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.85   11.85   clock network delay (propagated)
                   0.00   11.85   clock reconvergence pessimism
                          11.85 ^ core.CPU_Xreg_value_a4[2][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.17   11.68   library setup time
                          11.68   data required time
-----------------------------------------------------------------------------
                          11.68   data required time
                          -6.10   data arrival time
```

```
------------------------------------------------------------------------
                          5.58   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[16][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.29    0.30    4.71 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.12    0.13    4.85 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    1.16    0.98    5.82 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.20    0.28    6.10 v _10435_/Y (sky130_fd_sc_hd__o21ai_0)
           0.20    0.00    6.10 v core.CPU_Xreg_value_a4[16][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.10   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.85   11.85   clock network delay (propagated)
                   0.00   11.85   clock reconvergence pessimism
                          11.85 ^ core.CPU_Xreg_value_a4[16][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.17   11.68   library setup time
                          11.68   data required time
------------------------------------------------------------------------
                          11.68   data required time
                          -6.10   data arrival time
------------------------------------------------------------------------
                           5.58   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[3][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
```

```
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.29    0.30    4.71 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.12    0.13    4.85 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    1.16    0.98    5.82 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.19    0.27    6.09 v _09966_/Y (sky130_fd_sc_hd__o21ai_0)
           0.19    0.00    6.09 v core.CPU_Xreg_value_a4[3][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.09   data arrival time

                   11.00   11.00   clock clk (rise edge)
                    0.85   11.85   clock network delay (propagated)
                    0.00   11.85   clock reconvergence pessimism
                           11.85 ^ core.CPU_Xreg_value_a4[3][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.17   11.68   library setup time
                           11.68   data required time
-----------------------------------------------------------------------------
                           11.68   data required time
                           -6.09   data arrival time
-----------------------------------------------------------------------------
                            5.59   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
           0.07     0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13     0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36     0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16     0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16     0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36     0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15     0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97     0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25     0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14     0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.29     0.30    4.71 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.12     0.13    4.85 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    1.16     0.98    5.82 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.19     0.27    6.09 v _10712_/Y (sky130_fd_sc_hd__o21ai_0)
           0.19     0.00    6.09 v core.CPU_Xreg_value_a4[27][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                            6.09   data arrival time

                   11.00    11.00   clock clk (rise edge)
                    0.86    11.86   clock network delay (propagated)
                    0.00    11.86   clock reconvergence pessimism
                            11.86 ^ core.CPU_Xreg_value_a4[27][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.17    11.69   library setup time
                            11.69   data required time
-----------------------------------------------------------------------------
                            11.69   data required time
                            -6.09   data arrival time
-----------------------------------------------------------------------------
```

```
                          5.60   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[11][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.29    0.30    4.71 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.12    0.13    4.85 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    1.16    0.98    5.82 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.19    0.26    6.08 v _10339_/Y (sky130_fd_sc_hd__o21ai_0)
           0.19    0.00    6.08 v core.CPU_Xreg_value_a4[11][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.08   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.85   11.85   clock network delay (propagated)
                   0.00   11.85   clock reconvergence pessimism
                          11.85 ^ core.CPU_Xreg_value_a4[11][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.17   11.68   library setup time
                          11.68   data required time
---------------------------------------------------------------------------
                          11.68   data required time
                          -6.08   data arrival time
---------------------------------------------------------------------------
                           5.60   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[9][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
```

```
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.29    0.30    4.71 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.12    0.13    4.85 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    1.16    0.98    5.82 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.19    0.27    6.09 v _10158_/Y (sky130_fd_sc_hd__o21ai_0)
           0.19    0.00    6.09 v core.CPU_Xreg_value_a4[9][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.09   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.85   11.85   clock network delay (propagated)
                   0.00   11.85   clock reconvergence pessimism
                          11.85 ^ core.CPU_Xreg_value_a4[9][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.17   11.68   library setup time
                          11.68   data required time
-----------------------------------------------------------------------------
                          11.68   data required time
                          -6.09   data arrival time
-----------------------------------------------------------------------------
                           5.60   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[10][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.29    0.30    4.71 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.12    0.13    4.85 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    1.16    0.98    5.82 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.18    0.25    6.08 v _10247_/Y (sky130_fd_sc_hd__o21ai_0)
           0.18    0.00    6.08 v core.CPU_Xreg_value_a4[10][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.08   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.85   11.85   clock network delay (propagated)
                   0.00   11.85   clock reconvergence pessimism
                          11.85 ^ core.CPU_Xreg_value_a4[10][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.17   11.68   library setup time
                          11.68   data required time
-----------------------------------------------------------------------------
                          11.68   data required time
                          -6.08   data arrival time
-----------------------------------------------------------------------------
                           5.61   slack (MET)
```

```
Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.29    0.30    4.71 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.12    0.13    4.85 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    1.16    0.98    5.82 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.18    0.24    6.06 v _10524_/Y (sky130_fd_sc_hd__o21ai_0)
           0.18    0.00    6.06 v core.CPU_Xreg_value_a4[17][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.06   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.85   11.85   clock network delay (propagated)
                   0.00   11.85   clock reconvergence pessimism
                          11.85 ^ core.CPU_Xreg_value_a4[17][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.16   11.69   library setup time
                          11.69   data required time
---------------------------------------------------------------------
                          11.69   data required time
                          -6.06   data arrival time
---------------------------------------------------------------------
                           5.63   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[8][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
```

```
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.29    0.30    4.71 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.12    0.13    4.85 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    1.16    0.98    5.82 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.17    0.23    6.05 v _10066_/Y (sky130_fd_sc_hd__o21ai_0)
           0.17    0.00    6.05 v core.CPU_Xreg_value_a4[8][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.05   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.85   11.85   clock network delay (propagated)
                   0.00   11.85   clock reconvergence pessimism
                          11.85 ^ core.CPU_Xreg_value_a4[8][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.16   11.69   library setup time
                          11.69   data required time
-----------------------------------------------------------------------------
                          11.69   data required time
                          -6.05   data arrival time
-----------------------------------------------------------------------------
                           5.65   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[1][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01    0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.29    0.30    4.71 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.12    0.13    4.85 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    1.16    0.98    5.82 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.17    0.23    6.05 v _09692_/Y (sky130_fd_sc_hd__o21ai_0)
           0.17    0.00    6.05 v core.CPU_Xreg_value_a4[1][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           6.05   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.86   11.86   clock network delay (propagated)
                   0.00   11.86   clock reconvergence pessimism
                          11.86 ^ core.CPU_Xreg_value_a4[1][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.16   11.70   library setup time
                          11.70   data required time
-----------------------------------------------------------------------------
                          11.70   data required time
                          -6.05   data arrival time
-----------------------------------------------------------------------------
                           5.65   slack (MET)
```

```
Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_src2_value_a3[24]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
            0.07    0.00    0.97 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01     0.13    0.38    1.35 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13     0.36    0.36    1.71 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02     0.16    0.32    2.03 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01     0.16    0.28    2.31 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02     0.36    0.35    2.66 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01     0.15    0.21    2.87 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02     0.97    0.87    3.75 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.01     0.25    0.34    4.09 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03     0.14    0.32    4.41 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01     0.29    0.30    4.71 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01     0.12    0.13    4.85 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06     1.16    0.98    5.82 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00     0.19    0.27    6.09 v _07156_/Y (sky130_fd_sc_hd__o21ai_0)
            0.19    0.00    6.09 v core.CPU_src2_value_a3[24]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                            6.09   data arrival time

                   11.00   11.00   clock clk (rise edge)
                    0.98   11.98   clock network delay (propagated)
                    0.00   11.98   clock reconvergence pessimism
                           11.98 ^ core.CPU_src2_value_a3[24]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.17   11.81   library setup time
                           11.81   data required time
---------------------------------------------------------------------------
                           11.81   data required time
                           -6.09   data arrival time
---------------------------------------------------------------------------
                            5.72   slack (MET)


========== TT HOLD ==========
Startpoint: core.CPU_src2_value_a3[21]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_src2_value_a4[21]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.87    0.87   clock network delay (propagated)
            0.06    0.00    0.87 ^ core.CPU_src2_value_a3[21]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.02     0.08    0.35    1.22 v core.CPU_src2_value_a3[21]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
            0.08    0.00    1.22 v core.CPU_src2_value_a4[21]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                            1.22   data arrival time

                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
                    0.00    0.97   clock reconvergence pessimism
```

```
                          0.97 ^ core.CPU_src2_value_a4[21]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.06    0.91   library hold time
                          0.91   data required time
-----------------------------------------------------------------------
                          0.91   data required time
                         -1.22   data arrival time
-----------------------------------------------------------------------
                          0.31   slack (MET)


Startpoint: core.CPU_src2_value_a3[23]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_src2_value_a4[23]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.87    0.87   clock network delay (propagated)
           0.06    0.00    0.87 ^ core.CPU_src2_value_a3[23]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.02    0.08    0.35    1.22 v core.CPU_src2_value_a3[23]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.08    0.00    1.22 v core.CPU_src2_value_a4[23]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           1.22   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
                   0.00    0.97   clock reconvergence pessimism
                           0.97 ^ core.CPU_src2_value_a4[23]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.06    0.91   library hold time
                           0.91   data required time
-----------------------------------------------------------------------
                           0.91   data required time
                          -1.22   data arrival time
-----------------------------------------------------------------------
                           0.31   slack (MET)


Startpoint: core.CPU_rd_a2[4]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[4]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_rd_a2[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04    0.30    1.27 ^ core.CPU_rd_a2[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.04    0.00    1.27 ^ core.CPU_rd_a3[4]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           1.27   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
                   0.00    0.97   clock reconvergence pessimism
                           0.97 ^ core.CPU_rd_a3[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.94   library hold time
                           0.94   data required time
-----------------------------------------------------------------------
```

```
                          0.94   data required time
                         -1.27   data arrival time
---------------------------------------------------------------------------
                          0.33   slack (MET)


Startpoint: core.CPU_is_load_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_load_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
          0.07     0.00    0.97 ^ core.CPU_is_load_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04    0.30    1.27 ^ core.CPU_is_load_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.04     0.00    1.27 ^ core.CPU_is_load_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           1.27   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
                   0.00    0.97   clock reconvergence pessimism
                           0.97 ^ core.CPU_is_load_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.94   library hold time
                           0.94   data required time
---------------------------------------------------------------------------
                           0.94   data required time
                          -1.27   data arrival time
---------------------------------------------------------------------------
                           0.33   slack (MET)


Startpoint: core.CPU_rd_a2[3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[3]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
          0.07     0.00    0.97 ^ core.CPU_rd_a2[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04    0.31    1.27 ^ core.CPU_rd_a2[3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.04     0.00    1.27 ^ core.CPU_rd_a3[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           1.27   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
                   0.00    0.97   clock reconvergence pessimism
                           0.97 ^ core.CPU_rd_a3[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.94   library hold time
                           0.94   data required time
---------------------------------------------------------------------------
                           0.94   data required time
                          -1.27   data arrival time
---------------------------------------------------------------------------
                           0.33   slack (MET)
```

```
Startpoint: core.CPU_rd_a4[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a5[1]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
          0.07      0.00    0.97 ^ core.CPU_rd_a4[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00   0.04      0.31    1.28 ^ core.CPU_rd_a4[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.04      0.00    1.28 ^ core.CPU_rd_a5[1]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                            1.28   data arrival time

                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
                    0.00    0.97   clock reconvergence pessimism
                            0.97 ^ core.CPU_rd_a5[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.03    0.94   library hold time
                            0.94   data required time
---------------------------------------------------------------------
                            0.94   data required time
                           -1.28   data arrival time
---------------------------------------------------------------------
                            0.33   slack (MET)


Startpoint: core.CPU_inc_pc_a2[4]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[4]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
          0.07      0.00    0.97 ^ core.CPU_inc_pc_a2[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00   0.05      0.32    1.29 ^ core.CPU_inc_pc_a2[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.05      0.00    1.29 ^ core.CPU_inc_pc_a3[4]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                            1.29   data arrival time

                    0.00    0.00   clock clk (rise edge)
                    0.98    0.98   clock network delay (propagated)
                    0.00    0.98   clock reconvergence pessimism
                            0.98 ^ core.CPU_inc_pc_a3[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.03    0.95   library hold time
                            0.95   data required time
---------------------------------------------------------------------
                            0.95   data required time
                           -1.29   data arrival time
---------------------------------------------------------------------
                            0.34   slack (MET)


Startpoint: core.CPU_is_slli_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
```

```
Endpoint: core.CPU_is_slli_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.98    0.98   clock network delay (propagated)
           0.07    0.00    0.98 ^ core.CPU_is_slli_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04    0.31    1.29 ^ core.CPU_is_slli_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.04    0.00    1.29 ^ core.CPU_is_slli_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           1.29   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.98    0.98   clock network delay (propagated)
                   0.00    0.98   clock reconvergence pessimism
                           0.98 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.95   library hold time
                           0.95   data required time
---------------------------------------------------------------------------
                           0.95   data required time
                          -1.29   data arrival time
---------------------------------------------------------------------------
                           0.34   slack (MET)


Startpoint: core.CPU_rd_a4[4]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a5[4]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_rd_a4[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04    0.31    1.28 ^ core.CPU_rd_a4[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.04    0.00    1.28 ^ core.CPU_rd_a5[4]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           1.28   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
                   0.00    0.97   clock reconvergence pessimism
                           0.97 ^ core.CPU_rd_a5[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.94   library hold time
                           0.94   data required time
---------------------------------------------------------------------------
                           0.94   data required time
                          -1.28   data arrival time
---------------------------------------------------------------------------
                           0.34   slack (MET)


Startpoint: core.CPU_inc_pc_a2[5]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[5]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
```

```
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
           0.07     0.00    0.97 ^ core.CPU_inc_pc_a2[5]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04     0.31    1.28 ^ core.CPU_inc_pc_a2[5]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.04     0.00    1.28 ^ core.CPU_inc_pc_a3[5]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                            1.28   data arrival time

                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
                    0.00    0.97   clock reconvergence pessimism
                            0.97 ^ core.CPU_inc_pc_a3[5]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.03    0.94   library hold time
                            0.94   data required time
---------------------------------------------------------------------------
                            0.94   data required time
                           -1.28   data arrival time
---------------------------------------------------------------------------
                            0.34   slack (MET)


Startpoint: core.CPU_rd_a2[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[1]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
           0.07     0.00    0.97 ^ core.CPU_rd_a2[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.05     0.31    1.28 ^ core.CPU_rd_a2[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.05     0.00    1.28 ^ core.CPU_rd_a3[1]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                            1.28   data arrival time

                    0.00    0.00   clock clk (rise edge)
                    0.97    0.97   clock network delay (propagated)
                    0.00    0.97   clock reconvergence pessimism
                            0.97 ^ core.CPU_rd_a3[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.03    0.94   library hold time
                            0.94   data required time
---------------------------------------------------------------------------
                            0.94   data required time
                           -1.28   data arrival time
---------------------------------------------------------------------------
                            0.34   slack (MET)


Startpoint: core.CPU_inc_pc_a2[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[2]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
```

```
                  0.97    0.97   clock network delay (propagated)
          0.07    0.00    0.97 ^ core.CPU_inc_pc_a2[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.05    0.32    1.28 ^ core.CPU_inc_pc_a2[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.05    0.00    1.28 ^ core.CPU_inc_pc_a3[2]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                          1.28   data arrival time

                  0.00    0.00   clock clk (rise edge)
                  0.97    0.97   clock network delay (propagated)
                  0.00    0.97   clock reconvergence pessimism
                          0.97 ^ core.CPU_inc_pc_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.03    0.94   library hold time
                          0.94   data required time
-----------------------------------------------------------------------------
                          0.94   data required time
                         -1.28   data arrival time
-----------------------------------------------------------------------------
                          0.34   slack (MET)


Startpoint: core.CPU_is_addi_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_addi_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.97    0.97   clock network delay (propagated)
          0.07    0.00    0.97 ^ core.CPU_is_addi_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.05    0.32    1.29 ^ core.CPU_is_addi_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.05    0.00    1.29 ^ core.CPU_is_addi_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                          1.29   data arrival time

                  0.00    0.00   clock clk (rise edge)
                  0.97    0.97   clock network delay (propagated)
                  0.00    0.97   clock reconvergence pessimism
                          0.97 ^ core.CPU_is_addi_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.03    0.94   library hold time
                          0.94   data required time
---------------------------------------------------------------------------
                          0.94   data required time
                         -1.29   data arrival time
---------------------------------------------------------------------------
                          0.34   slack (MET)


Startpoint: core.CPU_is_beq_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_beq_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.97    0.97   clock network delay (propagated)
          0.07    0.00    0.97 ^ core.CPU_is_beq_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.05    0.31    1.29 ^ core.CPU_is_beq_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.05    0.00    1.29 ^ core.CPU_is_beq_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
```

```
                           1.29   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
                   0.00    0.97   clock reconvergence pessimism
                           0.97 ^ core.CPU_is_beq_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.94   library hold time
                           0.94   data required time
-----------------------------------------------------------------------------
                           0.94   data required time
                          -1.29   data arrival time
-----------------------------------------------------------------------------
                           0.34   slack (MET)


Startpoint: core.CPU_is_s_instr_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_s_instr_a4$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_is_s_instr_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.05    0.32    1.29 ^ core.CPU_is_s_instr_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.05    0.00    1.29 ^ core.CPU_is_s_instr_a4$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           1.29   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
                   0.00    0.97   clock reconvergence pessimism
                           0.97 ^ core.CPU_is_s_instr_a4$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.94   library hold time
                           0.94   data required time
---------------------------------------------------------------------------
                           0.94   data required time
                          -1.29   data arrival time
---------------------------------------------------------------------------
                           0.34   slack (MET)


Startpoint: core.CPU_rd_a4[3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_rd_a4[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.05    0.32    1.29 ^ core.CPU_rd_a4[3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.05    0.00    1.29 ^ core.CPU_rd_a5[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           1.29   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
```

```
                  0.00    0.97   clock reconvergence pessimism
                          0.97 ^ core.CPU_rd_a5[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.03    0.94   library hold time
                          0.94   data required time
-----------------------------------------------------------------------------
                          0.94   data required time
                         -1.29   data arrival time
-----------------------------------------------------------------------------
                          0.35   slack (MET)


Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.97    0.97   clock network delay (propagated)
           0.07   0.00    0.97 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.05   0.32    1.29 ^ core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.05   0.00    1.29 ^ core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                          1.29   data arrival time

                  0.00    0.00   clock clk (rise edge)
                  0.97    0.97   clock network delay (propagated)
                  0.00    0.97   clock reconvergence pessimism
                          0.97 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.03    0.94   library hold time
                          0.94   data required time
---------------------------------------------------------------------------
                          0.94   data required time
                         -1.29   data arrival time
---------------------------------------------------------------------------
                          0.35   slack (MET)


Startpoint: core.CPU_is_bne_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_bne_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.97    0.97   clock network delay (propagated)
           0.07   0.00    0.97 ^ core.CPU_is_bne_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.05   0.32    1.29 ^ core.CPU_is_bne_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.05   0.00    1.29 ^ core.CPU_is_bne_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                          1.29   data arrival time

                  0.00    0.00   clock clk (rise edge)
                  0.97    0.97   clock network delay (propagated)
                  0.00    0.97   clock reconvergence pessimism
                          0.97 ^ core.CPU_is_bne_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.03    0.94   library hold time
                          0.94   data required time
```

```
-------------------------------------------------------------------------
                          0.94   data required time
                         -1.29   data arrival time
-------------------------------------------------------------------------
                          0.35   slack (MET)


Startpoint: core.CPU_imm_a2[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_imm_a3[30]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_imm_a2[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.05    0.32    1.29 ^ core.CPU_imm_a2[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.05    0.00    1.29 ^ core.CPU_imm_a3[30]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           1.29   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
                   0.00    0.97   clock reconvergence pessimism
                           0.97 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.94   library hold time
                           0.94   data required time
-------------------------------------------------------------------------
                           0.94   data required time
                          -1.29   data arrival time
-------------------------------------------------------------------------
                           0.35   slack (MET)


Startpoint: core.CPU_is_sll_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_sll_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
-------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.98    0.98   clock network delay (propagated)
           0.07    0.00    0.98 ^ core.CPU_is_sll_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.05    0.32    1.30 ^ core.CPU_is_sll_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.05    0.00    1.30 ^ core.CPU_is_sll_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           1.30   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.98    0.98   clock network delay (propagated)
                   0.00    0.98   clock reconvergence pessimism
                           0.98 ^ core.CPU_is_sll_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.95   library hold time
                           0.95   data required time
-------------------------------------------------------------------------
                           0.95   data required time
                          -1.30   data arrival time
-------------------------------------------------------------------------
```

```
                          0.35   slack (MET)


Startpoint: core.CPU_rd_a4[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a5[0]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.07    0.00    0.97 ^ core.CPU_rd_a4[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.05    0.32    1.29 ^ core.CPU_rd_a4[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.05    0.00    1.29 ^ core.CPU_rd_a5[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           1.29   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
                   0.00    0.97   clock reconvergence pessimism
                           0.97 ^ core.CPU_rd_a5[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.94   library hold time
                           0.94   data required time
---------------------------------------------------------------------------
                           0.94   data required time
                          -1.29   data arrival time
---------------------------------------------------------------------------
                           0.35   slack (MET)


Startpoint: core.CPU_imm_a2[10]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_imm_a3[10]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
           0.06    0.00    0.97 ^ core.CPU_imm_a2[10]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.06    0.32    1.28 ^ core.CPU_imm_a2[10]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.06    0.00    1.29 ^ core.CPU_imm_a3[10]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           1.29   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
                   0.00    0.97   clock reconvergence pessimism
                           0.97 ^ core.CPU_imm_a3[10]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.93   library hold time
                           0.93   data required time
---------------------------------------------------------------------------
                           0.93   data required time
                          -1.29   data arrival time
---------------------------------------------------------------------------
                           0.35   slack (MET)


Startpoint: core.CPU_rd_valid_a2$_DFF_P_
```

```
           (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_valid_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
          0.07     0.00    0.97 ^ core.CPU_rd_valid_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.06    0.32    1.29 ^ core.CPU_rd_valid_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.06     0.00    1.29 ^ core.CPU_rd_valid_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           1.29   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.96    0.96   clock network delay (propagated)
                   0.00    0.96   clock reconvergence pessimism
                           0.96 ^ core.CPU_rd_valid_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.93   library hold time
                           0.93   data required time
---------------------------------------------------------------------------
                           0.93   data required time
                          -1.29   data arrival time
---------------------------------------------------------------------------
                           0.35   slack (MET)


Startpoint: core.CPU_inc_pc_a2[3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[3]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
          0.07     0.00    0.97 ^ core.CPU_inc_pc_a2[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.06    0.32    1.29 ^ core.CPU_inc_pc_a2[3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.06     0.00    1.29 ^ core.CPU_inc_pc_a3[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           1.29   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.97    0.97   clock network delay (propagated)
                   0.00    0.97   clock reconvergence pessimism
                           0.97 ^ core.CPU_inc_pc_a3[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.03    0.94   library hold time
                           0.94   data required time
---------------------------------------------------------------------------
                           0.94   data required time
                          -1.29   data arrival time
---------------------------------------------------------------------------
                           0.35   slack (MET)


Startpoint: core.CPU_Xreg_value_a5[17][1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.OUT[1]$_DFF_P_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
```

```
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.94    0.94   clock network delay (propagated)
           0.07     0.00    0.94 ^ core.CPU_Xreg_value_a5[17][1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04     0.31    1.25 v core.CPU_Xreg_value_a5[17][1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.04     0.00    1.25 v core.OUT[1]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                            1.25   data arrival time

                    0.00    0.00   clock clk (rise edge)
                    0.94    0.94   clock network delay (propagated)
                    0.00    0.94   clock reconvergence pessimism
                            0.94 ^ core.OUT[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.04    0.89   library hold time
                            0.89   data required time
---------------------------------------------------------------------------
                            0.89   data required time
                           -1.25   data arrival time
---------------------------------------------------------------------------
                            0.35   slack (MET)


========== TT WNS/TNS ==========
worst slack max 5.30
worst slack min 0.31
tns max 0.00
```

```
OpenROAD Static Timing Analysis Report
Design: vsdbabysoc | Library: sky130_fd_sc_hd
FF corner: design SDC clock, post-global-route (5_1_grt.odb), propagated clock

OpenROAD unknown
Features included (+) or not (-): -GPU +GUI -Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
========== FF SETUP ==========
Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_src2_value_a3[26]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
           0.06     0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10     0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28     0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13     0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12     0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25     0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10     0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67     0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18     0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10     0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.30     0.30    3.16 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09     0.09    3.24 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.08    0.90     0.73    3.98 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.13     0.08    4.06 v _07135_/Y (sky130_fd_sc_hd__o21ai_0)
           0.13     0.00    4.06 v core.CPU_src2_value_a3[26]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                            4.06   data arrival time

                   11.00   11.00   clock clk (rise edge)
                    0.61   11.61   clock network delay (propagated)
                    0.00   11.61   clock reconvergence pessimism
                           11.61 ^ core.CPU_src2_value_a3[26]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.10   11.50   library setup time
                           11.50   data required time
---------------------------------------------------------------------
                           11.50   data required time
                           -4.06   data arrival time
---------------------------------------------------------------------
                            7.45   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[1][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
           0.06     0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
```

```
   0.01    0.10    0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28    0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13    0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12    0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25    0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10    0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67    0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.30    0.30    3.16 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09    0.09    3.24 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.08    0.90    0.73    3.98 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.14    0.07    4.05 v _09682_/Y (sky130_fd_sc_hd__o221ai_1)
           0.14    0.00    4.05 v core.CPU_Xreg_value_a4[1][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           4.05   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.61   11.61   clock network delay (propagated)
                   0.00   11.61   clock reconvergence pessimism
                          11.61 ^ core.CPU_Xreg_value_a4[1][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.11   11.50   library setup time
                          11.50   data required time
-----------------------------------------------------------------------------
                          11.50   data required time
                          -4.05   data arrival time
-----------------------------------------------------------------------------
                           7.46   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[8][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.69    0.69   clock network delay (propagated)
           0.06    0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10    0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28    0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13    0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12    0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25    0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10    0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67    0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.30    0.30    3.16 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09    0.09    3.24 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.08    0.90    0.73    3.98 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.01    0.14    0.05    4.02 v _10060_/Y (sky130_fd_sc_hd__o221ai_1)
           0.14    0.00    4.02 v core.CPU_Xreg_value_a4[8][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           4.02   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.61   11.61   clock network delay (propagated)
                   0.00   11.61   clock reconvergence pessimism
                          11.61 ^ core.CPU_Xreg_value_a4[8][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.11   11.50   library setup time
```

```
                          11.50   data required time
-----------------------------------------------------------------------------
                          11.50   data required time
                          -4.02   data arrival time
-----------------------------------------------------------------------------
                           7.48   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
           0.06     0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10     0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28     0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13     0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12     0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25     0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10     0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67     0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18     0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10     0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.30     0.30    3.16 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09     0.09    3.24 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.08    0.90     0.73    3.98 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.01    0.14     0.04    4.02 v _10706_/Y (sky130_fd_sc_hd__o221ai_1)
           0.14     0.00    4.02 v core.CPU_Xreg_value_a4[27][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                            4.02   data arrival time

                   11.00   11.00   clock clk (rise edge)
                    0.61   11.61   clock network delay (propagated)
                    0.00   11.61   clock reconvergence pessimism
                           11.61 ^ core.CPU_Xreg_value_a4[27][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.11   11.50   library setup time
                           11.50   data required time
---------------------------------------------------------------------------
                           11.50   data required time
                           -4.02   data arrival time
---------------------------------------------------------------------------
                            7.48   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
           0.06     0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10     0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
```

```
   0.13    0.28    0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13    0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12    0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25    0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10    0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67    0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.30    0.30    3.16 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09    0.09    3.24 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.08    0.90    0.73    3.98 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.14    0.04    4.01 v _10518_/Y (sky130_fd_sc_hd__o221ai_1)
           0.14    0.00    4.01 v core.CPU_Xreg_value_a4[17][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           4.01   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.61   11.61   clock network delay (propagated)
                   0.00   11.61   clock reconvergence pessimism
                          11.61 ^ core.CPU_Xreg_value_a4[17][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.11   11.50   library setup time
                          11.50   data required time
-----------------------------------------------------------------------------
                          11.50   data required time
                          -4.01   data arrival time
-----------------------------------------------------------------------------
                           7.49   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[26][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap     Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
           0.06     0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10     0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28     0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13     0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12     0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25     0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10     0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67     0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18     0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10     0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.30     0.30    3.16 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09     0.09    3.24 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.08    0.90     0.73    3.98 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.14     0.03    4.01 v _10617_/Y (sky130_fd_sc_hd__o221ai_1)
           0.14     0.00    4.01 v core.CPU_Xreg_value_a4[26][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                            4.01   data arrival time

                   11.00   11.00   clock clk (rise edge)
                    0.61   11.61   clock network delay (propagated)
                    0.00   11.61   clock reconvergence pessimism
                           11.61 ^ core.CPU_Xreg_value_a4[26][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.11   11.50   library setup time
                           11.50   data required time
```

```
------------------------------------------------------------------------
                          11.50   data required time
                          -4.01   data arrival time
------------------------------------------------------------------------
                           7.50   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[3][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
           0.06     0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10     0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28     0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13     0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12     0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25     0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10     0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67     0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18     0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10     0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.30     0.30    3.16 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09     0.09    3.24 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.08    0.90     0.73    3.98 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.13     0.03    4.00 v _09959_/Y (sky130_fd_sc_hd__o221ai_1)
           0.13     0.00    4.00 v core.CPU_Xreg_value_a4[3][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                            4.00   data arrival time

                   11.00   11.00   clock clk (rise edge)
                    0.61   11.61   clock network delay (propagated)
                    0.00   11.61   clock reconvergence pessimism
                           11.61 ^ core.CPU_Xreg_value_a4[3][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.10   11.50   library setup time
                           11.50   data required time
------------------------------------------------------------------------
                          11.50   data required time
                          -4.00   data arrival time
------------------------------------------------------------------------
                           7.50   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
           0.06     0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10     0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28     0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
```

```
   0.02    0.13    0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12    0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25    0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10    0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67    0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.30    0.30    3.16 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09    0.09    3.24 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.08    0.90    0.73    3.98 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.13    0.03    4.00 v _09847_/Y (sky130_fd_sc_hd__o221ai_1)
           0.13    0.00    4.00 v core.CPU_Xreg_value_a4[2][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           4.00   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.61   11.61   clock network delay (propagated)
                   0.00   11.61   clock reconvergence pessimism
                          11.61 ^ core.CPU_Xreg_value_a4[2][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.10   11.50   library setup time
                          11.50   data required time
-----------------------------------------------------------------------------
                          11.50   data required time
                          -4.00   data arrival time
-----------------------------------------------------------------------------
                           7.50   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[16][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.69    0.69   clock network delay (propagated)
           0.06    0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10    0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28    0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13    0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12    0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25    0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10    0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67    0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.30    0.30    3.16 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09    0.09    3.24 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.08    0.90    0.73    3.98 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.13    0.02    4.00 v _10428_/Y (sky130_fd_sc_hd__o221ai_1)
           0.13    0.00    4.00 v core.CPU_Xreg_value_a4[16][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           4.00   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.61   11.61   clock network delay (propagated)
                   0.00   11.61   clock reconvergence pessimism
                          11.61 ^ core.CPU_Xreg_value_a4[16][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.10   11.50   library setup time
                          11.50   data required time
-----------------------------------------------------------------------------
```

```
                          11.50   data required time
                          -4.00   data arrival time
-----------------------------------------------------------------------------
                           7.50   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[9][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
          0.06      0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01     0.10    0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13     0.28    0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02     0.13    0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01     0.12    0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02     0.25    0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01     0.10    0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02     0.67    0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02     0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03     0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01     0.30    0.30    3.16 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01     0.09    0.09    3.24 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.08     0.90    0.73    3.98 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00     0.13    0.02    4.00 v _10151_/Y (sky130_fd_sc_hd__o221ai_1)
          0.13      0.00    4.00 v core.CPU_Xreg_value_a4[9][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                             4.00   data arrival time

                   11.00   11.00   clock clk (rise edge)
                    0.61   11.61   clock network delay (propagated)
                    0.00   11.61   clock reconvergence pessimism
                           11.61 ^ core.CPU_Xreg_value_a4[9][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.10   11.50   library setup time
                           11.50   data required time
-----------------------------------------------------------------------------
                          11.50   data required time
                          -4.00   data arrival time
-----------------------------------------------------------------------------
                           7.51   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[10][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
          0.06      0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01     0.10    0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13     0.28    0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02     0.13    0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
```

```
   0.01    0.12    0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25    0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10    0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67    0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.30    0.30    3.16 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09    0.09    3.24 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.08    0.90    0.73    3.98 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.13    0.02    3.99 v _10241_/Y (sky130_fd_sc_hd__o221ai_1)
           0.13    0.00    3.99 v core.CPU_Xreg_value_a4[10][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.99   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.61   11.61   clock network delay (propagated)
                   0.00   11.61   clock reconvergence pessimism
                          11.61 ^ core.CPU_Xreg_value_a4[10][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.10   11.50   library setup time
                          11.50   data required time
-----------------------------------------------------------------------------
                          11.50   data required time
                          -3.99   data arrival time
-----------------------------------------------------------------------------
                           7.51   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[11][26]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.69    0.69   clock network delay (propagated)
           0.06    0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10    0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28    0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13    0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12    0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25    0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10    0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67    0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.30    0.30    3.16 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09    0.09    3.24 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.08    0.90    0.73    3.98 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.12    0.01    3.98 v _10333_/Y (sky130_fd_sc_hd__o221ai_1)
           0.12    0.00    3.98 v core.CPU_Xreg_value_a4[11][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.98   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.61   11.61   clock network delay (propagated)
                   0.00   11.61   clock reconvergence pessimism
                          11.61 ^ core.CPU_Xreg_value_a4[11][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.10   11.51   library setup time
                          11.51   data required time
-----------------------------------------------------------------------------
                          11.51   data required time
```

```
                          -3.98   data arrival time
-----------------------------------------------------------------------------
                           7.52   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_src1_value_a3[26]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
           0.06     0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10     0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28     0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13     0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12     0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25     0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10     0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67     0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18     0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10     0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.30     0.30    3.16 ^ _06332_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09     0.09    3.24 v _06333_/Y (sky130_fd_sc_hd__xnor2_1)
   0.08    0.90     0.73    3.98 ^ _06373_/Y (sky130_fd_sc_hd__a211oi_2)
   0.00    0.17     0.01    3.99 v _06375_/Y (sky130_fd_sc_hd__mux2i_1)
           0.17     0.00    3.99 v core.CPU_src1_value_a3[26]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                             3.99   data arrival time

                   11.00    11.00   clock clk (rise edge)
                    0.69    11.69   clock network delay (propagated)
                    0.00    11.69   clock reconvergence pessimism
                            11.69 ^ core.CPU_src1_value_a3[26]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.12    11.57   library setup time
                            11.57   data required time
-----------------------------------------------------------------------------
                            11.57   data required time
                            -3.99   data arrival time
-----------------------------------------------------------------------------
                             7.58   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[26][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
           0.06     0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10     0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28     0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13     0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12     0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
```

```
   0.02    0.25    0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10    0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67    0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.20    0.20    3.06 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09    0.07    3.12 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    0.74    0.59    3.72 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.01    0.15    0.13    3.85 v _10623_/Y (sky130_fd_sc_hd__o21ai_0)
           0.15    0.00    3.85 v core.CPU_Xreg_value_a4[26][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.85   data arrival time

                   11.00   11.00   clock clk (rise edge)
                    0.61   11.61   clock network delay (propagated)
                    0.00   11.61   clock reconvergence pessimism
                           11.61 ^ core.CPU_Xreg_value_a4[26][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.11   11.50   library setup time
                           11.50   data required time
-----------------------------------------------------------------------------
                           11.50   data required time
                           -3.85   data arrival time
-----------------------------------------------------------------------------
                            7.65   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[2][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
           0.06     0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10     0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28     0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13     0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12     0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25     0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10     0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67     0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18     0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10     0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.20     0.20    3.06 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09     0.07    3.12 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    0.74     0.59    3.72 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.13     0.11    3.83 v _09855_/Y (sky130_fd_sc_hd__o21ai_0)
           0.13     0.00    3.83 v core.CPU_Xreg_value_a4[2][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                            3.83   data arrival time

                   11.00    11.00   clock clk (rise edge)
                    0.61    11.61   clock network delay (propagated)
                    0.00    11.61   clock reconvergence pessimism
                            11.61 ^ core.CPU_Xreg_value_a4[2][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.11    11.50   library setup time
                            11.50   data required time
---------------------------------------------------------------------------
                            11.50   data required time
                            -3.83   data arrival time
```

```
--------------------------------------------------------------------------
                          7.67   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[16][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
--------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.69    0.69   clock network delay (propagated)
           0.06    0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10    0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28    0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13    0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12    0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25    0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10    0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67    0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.20    0.20    3.06 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09    0.07    3.12 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    0.74    0.59    3.72 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.13    0.11    3.83 v _10435_/Y (sky130_fd_sc_hd__o21ai_0)
           0.13    0.00    3.83 v core.CPU_Xreg_value_a4[16][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.83   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.61   11.61   clock network delay (propagated)
                   0.00   11.61   clock reconvergence pessimism
                          11.61 ^ core.CPU_Xreg_value_a4[16][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.10   11.50   library setup time
                          11.50   data required time
--------------------------------------------------------------------------
                          11.50   data required time
                          -3.83   data arrival time
--------------------------------------------------------------------------
                           7.67   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[3][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
--------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.69    0.69   clock network delay (propagated)
           0.06    0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10    0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28    0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13    0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12    0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25    0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
```

```
   0.01    0.10    0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67    0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.20    0.20    3.06 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09    0.07    3.12 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    0.74    0.59    3.72 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.13    0.11    3.83 v _09966_/Y (sky130_fd_sc_hd__o21ai_0)
           0.13    0.00    3.83 v core.CPU_Xreg_value_a4[3][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.83   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.61   11.61   clock network delay (propagated)
                   0.00   11.61   clock reconvergence pessimism
                          11.61 ^ core.CPU_Xreg_value_a4[3][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.10   11.50   library setup time
                          11.50   data required time
-----------------------------------------------------------------------------
                          11.50   data required time
                          -3.83   data arrival time
-----------------------------------------------------------------------------
                           7.67   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[27][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.69    0.69   clock network delay (propagated)
           0.06    0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10    0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28    0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13    0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12    0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25    0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10    0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67    0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.20    0.20    3.06 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09    0.07    3.12 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    0.74    0.59    3.72 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.13    0.11    3.82 v _10712_/Y (sky130_fd_sc_hd__o21ai_0)
           0.13    0.00    3.83 v core.CPU_Xreg_value_a4[27][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.83   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.61   11.61   clock network delay (propagated)
                   0.00   11.61   clock reconvergence pessimism
                          11.61 ^ core.CPU_Xreg_value_a4[27][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.10   11.51   library setup time
                          11.51   data required time
-----------------------------------------------------------------------------
                          11.51   data required time
                          -3.83   data arrival time
-----------------------------------------------------------------------------
```

```
                          7.68   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[11][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.69    0.69   clock network delay (propagated)
          0.06     0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10    0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28    0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13    0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12    0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25    0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10    0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67    0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.20    0.20    3.06 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09    0.07    3.12 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    0.74    0.59    3.72 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.13    0.11    3.82 v _10339_/Y (sky130_fd_sc_hd__o21ai_0)
           0.13    0.00    3.82 v core.CPU_Xreg_value_a4[11][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.82   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.61   11.61   clock network delay (propagated)
                   0.00   11.61   clock reconvergence pessimism
                          11.61 ^ core.CPU_Xreg_value_a4[11][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.10   11.50   library setup time
                          11.50   data required time
---------------------------------------------------------------------------
                          11.50   data required time
                          -3.82   data arrival time
---------------------------------------------------------------------------
                           7.68   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[9][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.69    0.69   clock network delay (propagated)
          0.06     0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10    0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28    0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13    0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12    0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25    0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10    0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
```

```
   0.02    0.67    0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.20    0.20    3.06 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09    0.07    3.12 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    0.74    0.59    3.72 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.13    0.11    3.82 v _10158_/Y (sky130_fd_sc_hd__o21ai_0)
           0.13    0.00    3.82 v core.CPU_Xreg_value_a4[9][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.82   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.61   11.61   clock network delay (propagated)
                   0.00   11.61   clock reconvergence pessimism
                          11.61 ^ core.CPU_Xreg_value_a4[9][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.10   11.50   library setup time
                          11.50   data required time
-----------------------------------------------------------------------------
                          11.50   data required time
                          -3.82   data arrival time
-----------------------------------------------------------------------------
                           7.68   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[10][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.69    0.69   clock network delay (propagated)
           0.06    0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10    0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28    0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13    0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12    0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25    0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10    0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67    0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.20    0.20    3.06 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09    0.07    3.12 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    0.74    0.59    3.72 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.13    0.10    3.82 v _10247_/Y (sky130_fd_sc_hd__o21ai_0)
           0.13    0.00    3.82 v core.CPU_Xreg_value_a4[10][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.82   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.61   11.61   clock network delay (propagated)
                   0.00   11.61   clock reconvergence pessimism
                          11.61 ^ core.CPU_Xreg_value_a4[10][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.10   11.50   library setup time
                          11.50   data required time
-----------------------------------------------------------------------------
                          11.50   data required time
                          -3.82   data arrival time
-----------------------------------------------------------------------------
                           7.69   slack (MET)
```

```
Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[17][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
           0.06     0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10     0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28     0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13     0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12     0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25     0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10     0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67     0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18     0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10     0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.20     0.20    3.06 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09     0.07    3.12 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    0.74     0.59    3.72 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.12     0.09    3.81 v _10524_/Y (sky130_fd_sc_hd__o21ai_0)
           0.12     0.00    3.81 v core.CPU_Xreg_value_a4[17][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                             3.81   data arrival time

                   11.00    11.00   clock clk (rise edge)
                    0.61    11.61   clock network delay (propagated)
                    0.00    11.61   clock reconvergence pessimism
                            11.61 ^ core.CPU_Xreg_value_a4[17][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.10    11.51   library setup time
                            11.51   data required time
---------------------------------------------------------------------
                            11.51   data required time
                            -3.81   data arrival time
---------------------------------------------------------------------
                             7.70   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[8][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
           0.06     0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10     0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28     0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13     0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12     0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25     0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10     0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67     0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
```

```
   0.02    0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.20    0.20    3.06 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09    0.07    3.12 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    0.74    0.59    3.72 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.12    0.08    3.80 v _10066_/Y (sky130_fd_sc_hd__o21ai_0)
           0.12    0.00    3.80 v core.CPU_Xreg_value_a4[8][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.80   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.61   11.61   clock network delay (propagated)
                   0.00   11.61   clock reconvergence pessimism
                          11.61 ^ core.CPU_Xreg_value_a4[8][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.10   11.51   library setup time
                          11.51   data required time
-----------------------------------------------------------------------------
                          11.51   data required time
                          -3.80   data arrival time
-----------------------------------------------------------------------------
                           7.71   slack (MET)


Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[1][24]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.69    0.69   clock network delay (propagated)
           0.06    0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10    0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28    0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13    0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12    0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25    0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10    0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67    0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18    0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.03    0.10    0.21    2.86 v _06201_/X (sky130_fd_sc_hd__and2_1)
   0.01    0.20    0.20    3.06 ^ _06439_/Y (sky130_fd_sc_hd__o31ai_1)
   0.01    0.09    0.07    3.12 v _06440_/Y (sky130_fd_sc_hd__xnor2_1)
   0.06    0.74    0.59    3.72 ^ _06458_/Y (sky130_fd_sc_hd__a221oi_2)
   0.00    0.12    0.08    3.80 v _09692_/Y (sky130_fd_sc_hd__o21ai_0)
           0.12    0.00    3.80 v core.CPU_Xreg_value_a4[1][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                           3.80   data arrival time

                  11.00   11.00   clock clk (rise edge)
                   0.61   11.61   clock network delay (propagated)
                   0.00   11.61   clock reconvergence pessimism
                          11.61 ^ core.CPU_Xreg_value_a4[1][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.10   11.51   library setup time
                          11.51   data required time
---------------------------------------------------------------------------
                          11.51   data required time
                          -3.80   data arrival time
---------------------------------------------------------------------------
                           7.71   slack (MET)
```

```
Startpoint: core.CPU_src2_value_a3[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_Xreg_value_a4[11][30]$_SDFFE_PP0P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
           0.06     0.00    0.69 ^ core.CPU_src2_value_a3[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.01    0.10     0.24    0.93 ^ core.CPU_src2_value_a3[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.13    0.28     0.26    1.19 ^ place387/X (sky130_fd_sc_hd__buf_4)
   0.02    0.13     0.20    1.39 ^ _11028_/COUT (sky130_fd_sc_hd__ha_1)
   0.01    0.12     0.18    1.57 ^ _10976_/COUT (sky130_fd_sc_hd__fa_1)
   0.02    0.25     0.23    1.81 ^ _05732_/Y (sky130_fd_sc_hd__nor2b_1)
   0.01    0.10     0.11    1.92 v _05736_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.67     0.57    2.49 ^ _05744_/Y (sky130_fd_sc_hd__a2111oi_1)
   0.02    0.18     0.16    2.65 v _05754_/Y (sky130_fd_sc_hd__o211ai_1)
   0.02    0.40     0.38    3.03 ^ _05777_/Y (sky130_fd_sc_hd__a211oi_1)
   0.00    0.04     0.10    3.12 ^ _05778_/X (sky130_fd_sc_hd__or4_1)
   0.01    0.10     0.07    3.20 v _05893_/Y (sky130_fd_sc_hd__a32oi_1)
   0.04    0.66     0.54    3.74 ^ _09640_/Y (sky130_fd_sc_hd__o221ai_1)
   0.01    0.13     0.12    3.86 v _10316_/Y (sky130_fd_sc_hd__o21ai_0)
           0.13     0.00    3.86 v core.CPU_Xreg_value_a4[11][30]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
                            3.86   data arrival time

                   11.00   11.00   clock clk (rise edge)
                    0.69   11.69   clock network delay (propagated)
                    0.00   11.69   clock reconvergence pessimism
                           11.69 ^ core.CPU_Xreg_value_a4[11][30]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.10   11.58   library setup time
                           11.58   data required time
---------------------------------------------------------------------------
                           11.58   data required time
                           -3.86   data arrival time
---------------------------------------------------------------------------
                            7.72   slack (MET)


========== FF HOLD ==========
Startpoint: core.CPU_src2_value_a3[21]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_src2_value_a4[21]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.62    0.62   clock network delay (propagated)
           0.05     0.00    0.62 ^ core.CPU_src2_value_a3[21]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.02    0.06     0.23    0.84 v core.CPU_src2_value_a3[21]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.06     0.00    0.85 v core.CPU_src2_value_a4[21]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                            0.85   data arrival time

                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
                    0.00    0.69   clock reconvergence pessimism
```

```
                          0.69 ^ core.CPU_src2_value_a4[21]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.05    0.64   library hold time
                          0.64   data required time
-----------------------------------------------------------------------------
                          0.64   data required time
                         -0.85   data arrival time
-----------------------------------------------------------------------------
                          0.20   slack (MET)


Startpoint: core.CPU_src2_value_a3[23]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_src2_value_a4[23]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.62    0.62   clock network delay (propagated)
           0.05    0.00    0.62 ^ core.CPU_src2_value_a3[23]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.02    0.06    0.23    0.84 v core.CPU_src2_value_a3[23]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.06    0.00    0.85 v core.CPU_src2_value_a4[23]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.85   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.69    0.69   clock network delay (propagated)
                   0.00    0.69   clock reconvergence pessimism
                           0.69 ^ core.CPU_src2_value_a4[23]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.05    0.64   library hold time
                           0.64   data required time
-----------------------------------------------------------------------------
                           0.64   data required time
                          -0.85   data arrival time
-----------------------------------------------------------------------------
                           0.20   slack (MET)


Startpoint: core.CPU_rd_a2[4]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[4]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
           0.05    0.00    0.68 ^ core.CPU_rd_a2[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.87 ^ core.CPU_rd_a2[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.87 ^ core.CPU_rd_a3[4]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.87   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
                   0.00    0.68   clock reconvergence pessimism
                           0.68 ^ core.CPU_rd_a3[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.02    0.66   library hold time
                           0.66   data required time
-----------------------------------------------------------------------------
```

```
                          0.66   data required time
                         -0.87   data arrival time
-----------------------------------------------------------------------------
                          0.21   slack (MET)


Startpoint: core.CPU_is_load_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_load_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
          0.05     0.00    0.68 ^ core.CPU_is_load_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.87 ^ core.CPU_is_load_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.03     0.00    0.87 ^ core.CPU_is_load_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.87   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
                   0.00    0.68   clock reconvergence pessimism
                           0.68 ^ core.CPU_is_load_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.02    0.66   library hold time
                           0.66   data required time
---------------------------------------------------------------------------
                           0.66   data required time
                          -0.87   data arrival time
---------------------------------------------------------------------------
                           0.21   slack (MET)


Startpoint: core.CPU_rd_a2[3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[3]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
          0.05     0.00    0.68 ^ core.CPU_rd_a2[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.87 ^ core.CPU_rd_a2[3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.03     0.00    0.87 ^ core.CPU_rd_a3[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.87   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
                   0.00    0.68   clock reconvergence pessimism
                           0.68 ^ core.CPU_rd_a3[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.02    0.66   library hold time
                           0.66   data required time
---------------------------------------------------------------------------
                           0.66   data required time
                          -0.87   data arrival time
---------------------------------------------------------------------------
                           0.21   slack (MET)
```

```
Startpoint: core.CPU_rd_a4[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a5[1]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.68    0.68   clock network delay (propagated)
          0.06      0.00    0.68 ^ core.CPU_rd_a4[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00   0.03      0.19    0.87 ^ core.CPU_rd_a4[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.03      0.00    0.87 ^ core.CPU_rd_a5[1]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                            0.87   data arrival time

                    0.00    0.00   clock clk (rise edge)
                    0.68    0.68   clock network delay (propagated)
                    0.00    0.68   clock reconvergence pessimism
                            0.68 ^ core.CPU_rd_a5[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.02    0.67   library hold time
                            0.67   data required time
---------------------------------------------------------------------
                            0.67   data required time
                           -0.87   data arrival time
---------------------------------------------------------------------
                            0.21   slack (MET)


Startpoint: core.CPU_inc_pc_a2[4]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[4]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.68    0.68   clock network delay (propagated)
          0.05      0.00    0.68 ^ core.CPU_inc_pc_a2[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00   0.04      0.20    0.88 ^ core.CPU_inc_pc_a2[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.04      0.00    0.88 ^ core.CPU_inc_pc_a3[4]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                            0.88   data arrival time

                    0.00    0.00   clock clk (rise edge)
                    0.69    0.69   clock network delay (propagated)
                    0.00    0.69   clock reconvergence pessimism
                            0.69 ^ core.CPU_inc_pc_a3[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.02    0.67   library hold time
                            0.67   data required time
---------------------------------------------------------------------
                            0.67   data required time
                           -0.88   data arrival time
---------------------------------------------------------------------
                            0.21   slack (MET)


Startpoint: core.CPU_is_slli_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
```

```
Endpoint: core.CPU_is_slli_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.69    0.69   clock network delay (propagated)
           0.06    0.00    0.69 ^ core.CPU_is_slli_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.88 ^ core.CPU_is_slli_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.88 ^ core.CPU_is_slli_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.88   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.69    0.69   clock network delay (propagated)
                   0.00    0.69   clock reconvergence pessimism
                           0.69 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.02    0.67   library hold time
                           0.67   data required time
---------------------------------------------------------------------------
                           0.67   data required time
                          -0.88   data arrival time
---------------------------------------------------------------------------
                           0.21   slack (MET)


Startpoint: core.CPU_rd_a4[4]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a5[4]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
           0.06    0.00    0.68 ^ core.CPU_rd_a4[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.19    0.88 ^ core.CPU_rd_a4[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.88 ^ core.CPU_rd_a5[4]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.88   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
                   0.00    0.68   clock reconvergence pessimism
                           0.68 ^ core.CPU_rd_a5[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.02    0.67   library hold time
                           0.67   data required time
---------------------------------------------------------------------------
                           0.67   data required time
                          -0.88   data arrival time
---------------------------------------------------------------------------
                           0.21   slack (MET)


Startpoint: core.CPU_inc_pc_a2[5]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[5]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
```

```
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.68    0.68   clock network delay (propagated)
           0.05     0.00    0.68 ^ core.CPU_inc_pc_a2[5]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03     0.19    0.88 ^ core.CPU_inc_pc_a2[5]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03     0.00    0.88 ^ core.CPU_inc_pc_a3[5]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                            0.88   data arrival time

                    0.00    0.00   clock clk (rise edge)
                    0.68    0.68   clock network delay (propagated)
                    0.00    0.68   clock reconvergence pessimism
                            0.68 ^ core.CPU_inc_pc_a3[5]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.02    0.67   library hold time
                            0.67   data required time
---------------------------------------------------------------------------
                            0.67   data required time
                           -0.88   data arrival time
---------------------------------------------------------------------------
                            0.21   slack (MET)


Startpoint: core.CPU_rd_a2[1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a3[1]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.68    0.68   clock network delay (propagated)
           0.05     0.00    0.68 ^ core.CPU_rd_a2[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03     0.19    0.87 ^ core.CPU_rd_a2[1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03     0.00    0.87 ^ core.CPU_rd_a3[1]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                            0.87   data arrival time

                    0.00    0.00   clock clk (rise edge)
                    0.68    0.68   clock network delay (propagated)
                    0.00    0.68   clock reconvergence pessimism
                            0.68 ^ core.CPU_rd_a3[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.02    0.66   library hold time
                            0.66   data required time
---------------------------------------------------------------------------
                            0.66   data required time
                           -0.87   data arrival time
---------------------------------------------------------------------------
                            0.21   slack (MET)


Startpoint: core.CPU_inc_pc_a2[2]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[2]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
```

```
                  0.68    0.68   clock network delay (propagated)
           0.05    0.00    0.68 ^ core.CPU_inc_pc_a2[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04    0.20    0.88 ^ core.CPU_inc_pc_a2[2]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.04    0.00    0.88 ^ core.CPU_inc_pc_a3[2]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.88   data arrival time

                  0.00    0.00   clock clk (rise edge)
                  0.68    0.68   clock network delay (propagated)
                  0.00    0.68   clock reconvergence pessimism
                          0.68 ^ core.CPU_inc_pc_a3[2]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.02    0.67   library hold time
                          0.67   data required time
-----------------------------------------------------------------------------
                          0.67   data required time
                         -0.88   data arrival time
-----------------------------------------------------------------------------
                          0.21   slack (MET)


Startpoint: core.CPU_is_addi_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_addi_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.69    0.69   clock network delay (propagated)
           0.06    0.00    0.69 ^ core.CPU_is_addi_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.20    0.88 ^ core.CPU_is_addi_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.88 ^ core.CPU_is_addi_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.88   data arrival time

                  0.00    0.00   clock clk (rise edge)
                  0.69    0.69   clock network delay (propagated)
                  0.00    0.69   clock reconvergence pessimism
                          0.69 ^ core.CPU_is_addi_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.02    0.67   library hold time
                          0.67   data required time
---------------------------------------------------------------------------
                          0.67   data required time
                         -0.88   data arrival time
---------------------------------------------------------------------------
                          0.21   slack (MET)


Startpoint: core.CPU_is_beq_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_beq_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.69    0.69   clock network delay (propagated)
           0.06    0.00    0.69 ^ core.CPU_is_beq_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.03    0.20    0.88 ^ core.CPU_is_beq_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.03    0.00    0.88 ^ core.CPU_is_beq_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
```

```
                          0.88   data arrival time

                  0.00    0.00   clock clk (rise edge)
                  0.68    0.68   clock network delay (propagated)
                  0.00    0.68   clock reconvergence pessimism
                          0.68 ^ core.CPU_is_beq_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.02    0.67   library hold time
                          0.67   data required time
-----------------------------------------------------------------------------
                          0.67   data required time
                         -0.88   data arrival time
-----------------------------------------------------------------------------
                          0.21   slack (MET)


Startpoint: core.CPU_is_s_instr_a3$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_s_instr_a4$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.69    0.69   clock network delay (propagated)
          0.05    0.00    0.69 ^ core.CPU_is_s_instr_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04    0.20    0.89 ^ core.CPU_is_s_instr_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.04    0.00    0.89 ^ core.CPU_is_s_instr_a4$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                          0.89   data arrival time

                  0.00    0.00   clock clk (rise edge)
                  0.69    0.69   clock network delay (propagated)
                  0.00    0.69   clock reconvergence pessimism
                          0.69 ^ core.CPU_is_s_instr_a4$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.02    0.67   library hold time
                          0.67   data required time
---------------------------------------------------------------------------
                          0.67   data required time
                         -0.89   data arrival time
---------------------------------------------------------------------------
                          0.21   slack (MET)


Startpoint: core.CPU_rd_a4[3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a5[3]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.68    0.68   clock network delay (propagated)
          0.06    0.00    0.68 ^ core.CPU_rd_a4[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04    0.20    0.88 ^ core.CPU_rd_a4[3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.04    0.00    0.88 ^ core.CPU_rd_a5[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                          0.88   data arrival time

                  0.00    0.00   clock clk (rise edge)
                  0.68    0.68   clock network delay (propagated)
```

```
                  0.00    0.68   clock reconvergence pessimism
                          0.68 ^ core.CPU_rd_a5[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.02    0.67   library hold time
                          0.67   data required time
-----------------------------------------------------------------------------
                          0.67   data required time
                         -0.88   data arrival time
-----------------------------------------------------------------------------
                          0.22   slack (MET)


Startpoint: core.CPU_reset_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_reset_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.69    0.69   clock network delay (propagated)
           0.06   0.00    0.69 ^ core.CPU_reset_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04   0.20    0.88 ^ core.CPU_reset_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.04   0.00    0.88 ^ core.CPU_reset_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                          0.88   data arrival time

                  0.00    0.00   clock clk (rise edge)
                  0.69    0.69   clock network delay (propagated)
                  0.00    0.69   clock reconvergence pessimism
                          0.69 ^ core.CPU_reset_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.02    0.67   library hold time
                          0.67   data required time
---------------------------------------------------------------------------
                          0.67   data required time
                         -0.88   data arrival time
---------------------------------------------------------------------------
                          0.22   slack (MET)


Startpoint: core.CPU_is_bne_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_bne_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00   clock clk (rise edge)
                  0.68    0.68   clock network delay (propagated)
           0.06   0.00    0.68 ^ core.CPU_is_bne_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04   0.20    0.88 ^ core.CPU_is_bne_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.04   0.00    0.88 ^ core.CPU_is_bne_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                          0.88   data arrival time

                  0.00    0.00   clock clk (rise edge)
                  0.68    0.68   clock network delay (propagated)
                  0.00    0.68   clock reconvergence pessimism
                          0.68 ^ core.CPU_is_bne_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 -0.02    0.67   library hold time
                          0.67   data required time
```

```
--------------------------------------------------------------------------
                          0.67   data required time
                         -0.88   data arrival time
--------------------------------------------------------------------------
                          0.22   slack (MET)


Startpoint: core.CPU_imm_a2[30]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_imm_a3[30]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
--------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.69    0.69   clock network delay (propagated)
           0.06    0.00    0.69 ^ core.CPU_imm_a2[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04    0.20    0.88 ^ core.CPU_imm_a2[30]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.04    0.00    0.88 ^ core.CPU_imm_a3[30]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.88   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
                   0.00    0.68   clock reconvergence pessimism
                           0.68 ^ core.CPU_imm_a3[30]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.02    0.67   library hold time
                           0.67   data required time
--------------------------------------------------------------------------
                           0.67   data required time
                          -0.88   data arrival time
--------------------------------------------------------------------------
                           0.22   slack (MET)


Startpoint: core.CPU_is_sll_a2$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_is_sll_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
--------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.69    0.69   clock network delay (propagated)
           0.06    0.00    0.69 ^ core.CPU_is_sll_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04    0.20    0.89 ^ core.CPU_is_sll_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.04    0.00    0.89 ^ core.CPU_is_sll_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.89   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.69    0.69   clock network delay (propagated)
                   0.00    0.69   clock reconvergence pessimism
                           0.69 ^ core.CPU_is_sll_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.02    0.67   library hold time
                           0.67   data required time
--------------------------------------------------------------------------
                           0.67   data required time
                          -0.89   data arrival time
--------------------------------------------------------------------------
```

```
                          0.22   slack (MET)


Startpoint: core.CPU_rd_a4[0]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_a5[0]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
           0.06    0.00    0.68 ^ core.CPU_rd_a4[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04    0.20    0.88 ^ core.CPU_rd_a4[0]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.04    0.00    0.88 ^ core.CPU_rd_a5[0]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.88   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
                   0.00    0.68   clock reconvergence pessimism
                           0.68 ^ core.CPU_rd_a5[0]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.02    0.67   library hold time
                           0.67   data required time
---------------------------------------------------------------------------
                           0.67   data required time
                          -0.88   data arrival time
---------------------------------------------------------------------------
                           0.22   slack (MET)


Startpoint: core.CPU_imm_a2[10]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_imm_a3[10]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
           0.05    0.00    0.68 ^ core.CPU_imm_a2[10]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04    0.20    0.88 ^ core.CPU_imm_a2[10]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.04    0.00    0.88 ^ core.CPU_imm_a3[10]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.88   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
                   0.00    0.68   clock reconvergence pessimism
                           0.68 ^ core.CPU_imm_a3[10]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.02    0.66   library hold time
                           0.66   data required time
---------------------------------------------------------------------------
                           0.66   data required time
                          -0.88   data arrival time
---------------------------------------------------------------------------
                           0.22   slack (MET)


Startpoint: core.CPU_rd_valid_a2$_DFF_P_
```

```
           (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_rd_valid_a3$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
          0.05     0.00    0.68 ^ core.CPU_rd_valid_a2$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04    0.20    0.88 ^ core.CPU_rd_valid_a2$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.04     0.00    0.88 ^ core.CPU_rd_valid_a3$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.88   data arrival time

                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
                   0.00    0.68   clock reconvergence pessimism
                           0.68 ^ core.CPU_rd_valid_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.02    0.66   library hold time
                           0.66   data required time
---------------------------------------------------------------------------
                           0.66   data required time
                          -0.88   data arrival time
---------------------------------------------------------------------------
                           0.22   slack (MET)


Startpoint: core.CPU_inc_pc_a2[3]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.CPU_inc_pc_a3[3]$_DFF_P_
          (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
          0.05     0.00    0.68 ^ core.CPU_inc_pc_a2[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04    0.20    0.88 ^ core.CPU_inc_pc_a2[3]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
          0.04     0.00    0.88 ^ core.CPU_inc_pc_a3[3]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                           0.88   data arrival time

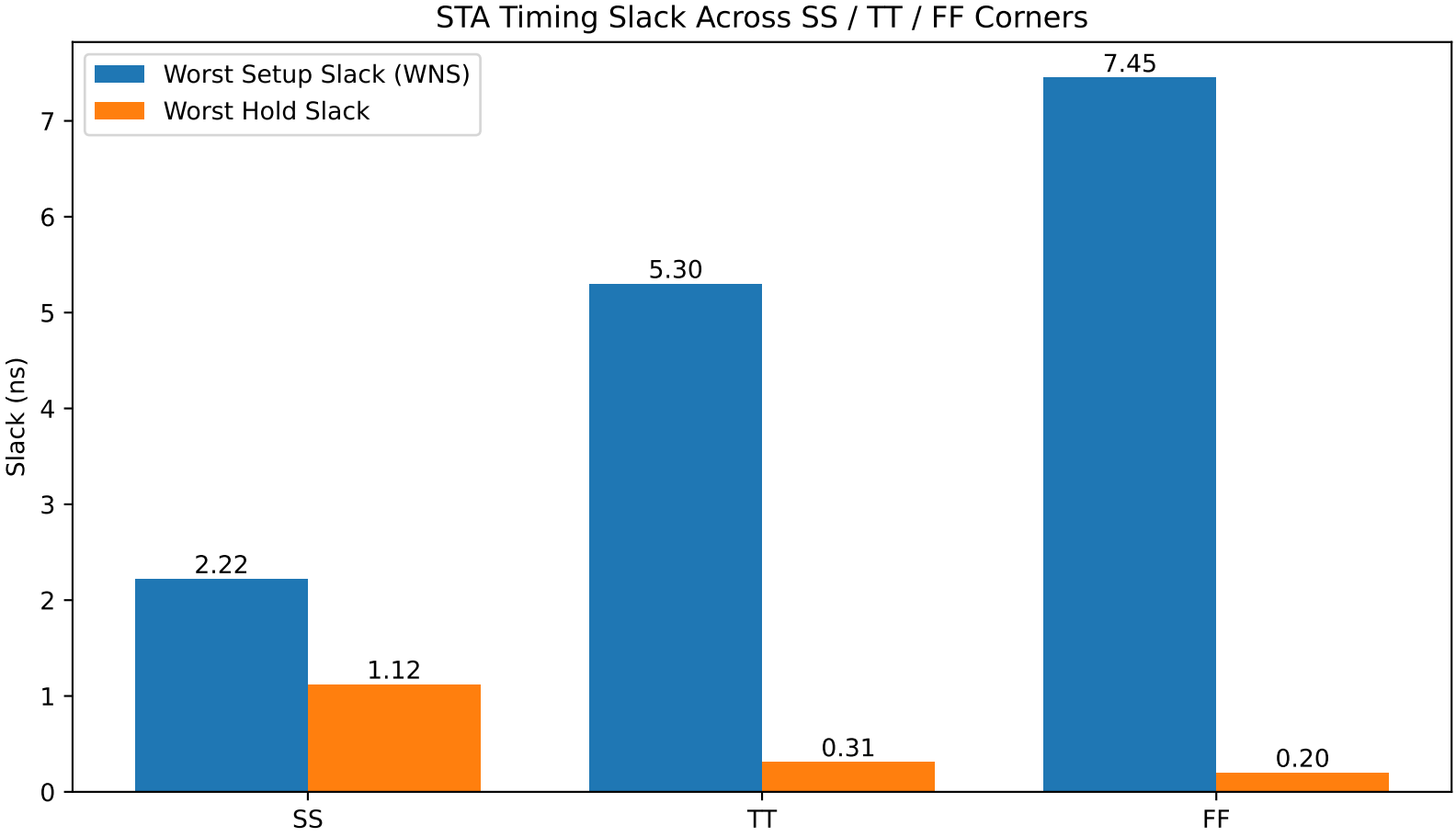
                   0.00    0.00   clock clk (rise edge)
                   0.68    0.68   clock network delay (propagated)
                   0.00    0.68   clock reconvergence pessimism
                           0.68 ^ core.CPU_inc_pc_a3[3]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  -0.02    0.67   library hold time
                           0.67   data required time
---------------------------------------------------------------------------
                           0.67   data required time
                          -0.88   data arrival time
---------------------------------------------------------------------------
                           0.22   slack (MET)


Startpoint: core.CPU_Xreg_value_a5[17][1]$_DFF_P_
            (rising edge-triggered flip-flop clocked by clk)
Endpoint: core.OUT[1]$_DFF_P_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
```

```
    Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                    0.00    0.00   clock clk (rise edge)
                    0.66    0.66   clock network delay (propagated)
           0.05     0.00    0.66 ^ core.CPU_Xreg_value_a5[17][1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.00    0.04     0.20    0.86 ^ core.CPU_Xreg_value_a5[17][1]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
           0.04     0.00    0.86 ^ core.OUT[1]$_DFF_P_/D (sky130_fd_sc_hd__dfxtp_1)
                            0.86   data arrival time

                    0.00    0.00   clock clk (rise edge)
                    0.66    0.66   clock network delay (propagated)
                    0.00    0.66   clock reconvergence pessimism
                            0.66 ^ core.OUT[1]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
                   -0.02    0.65   library hold time
                            0.65   data required time
---------------------------------------------------------------------------
                            0.65   data required time
                           -0.86   data arrival time
---------------------------------------------------------------------------
                            0.22   slack (MET)


========== FF WNS/TNS ==========
worst slack max 7.45
worst slack min 0.20
tns max 0.00
```


STA Timing Slack Across SS / TT / FF Corners
Worst Setup Slack (WNS)
Worst Hold Slack
Slack (ns)
0
1
2
3
4
5
6
7
2.22
1.12
5.30
0.31
7.45
0.20
SS
TT
FF